Automated Flashlamp Microscopic Flaw Inspections Group

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19 May 2026

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Bruno & Evan,

We have attached our report, Design of an Automated Flashlamp Microscopic Flaw Inspection System. This report outlines our design solution which has successfully met all requirements involved in the inspection of flashlamps at NIF.

The purpose of this report is to document our design solution by enumerating decisions made during conceptual design, detailed design and fabrication, such as our custom components and purchased parts.

It has been an honor and a privilege working with you on this project and we hope that you are able to make good use of it in future flashlamp inspections.

Sincerely,

The Automated Flashlamp Microscopic Flaw Inspection Group

Design of an Automated Flashlamp Flaw Inspection System

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EXECUTIVE SUMMARY

Lawrence Livermore National Laboratory (LLNL) is a national laboratory which is focused on enhancing national security through nuclear deterrence and energy security. The National Ignition Facility (NIF) is a research facility located at LLNL which is focused on high energy fusion research. A critical component in the energy transfer required to create a high energy laser beam is a flashlamp. Flashlamps are tube shaped lamps which emit white light to amplify the laser. Thousands of these lamps are used at the NIF during each experiment, significant defects will lead to catastrophe. Currently, flashlamps are manually inspected taking multiple hours per lamp.

The Automated Flashlamp Microscopic Flaw Inspection project has developed a system, in the figure shown below, which automates the process of inspecting flashlamps. A camera system takes resolute images of the entire surface area of the flashlamp, then feeds these images to a machine vision program, which identifies size, location and type of defect. The system images the flashlamp with two primary degrees of freedom: rotation of the flashlamp with a rotary table and longitudinal translation of the camera along a ball screw. A manually adjusted secondary camera is used to image the endcaps of flashlamps. Together, each of these components aid in imaging the entire surface of the flashlamp, meeting project requirements.

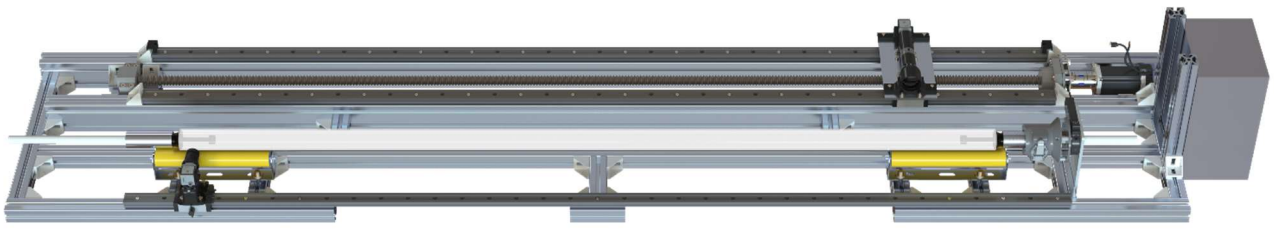


Figure. Design solution.

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NOMENCLATURE

A – Contact area between glass and polyurethane

CAD – Computer Aided Design

d – Outer flashlamp diameter

$\boldsymbol{d_i}$ **–** Inner flashlamp diameter

$\boldsymbol{d_o}$ **–** Clamp diameter

DT - Deuterium-Tritium

$\boldsymbol{E_i}$ **–** Young's modulus of glass

$\boldsymbol{E_o}$ **–** Young's modulus of polyurethane

Flashlamp – Gas filled glass tube with electrodes at either end, emits light when charged

FPS – Framerate of camera

HFOV – Horizontal field of view of lens

I – Moment of inertia of flashlamp

KDP - Potassium-Dihydrogen-Phosphate

$\boldsymbol{l}$ **–** Length of flashlamp observed by camera

LLNL – Lawrence Livermore National Laboratory

M – Magnification of telecentric lens

$\boldsymbol{N_{pixel}}$ **–** Pixel blur

NIF - National Ignition Facility

p – Pixel size of camera sensor

PC – Polycarbonate

PEEK – Polyether ether ketone

Pugh Analysis – Method of numerically comparing different designs against each other

$\boldsymbol{t_{exp}}$ **–** Exposure time of camera

Tube – Flashlamp without electrodes

UV – Ultraviolet

$\boldsymbol{v_b}$ **–** Maximum velocity of ball screw

VFOV – Vertical field of view of lens

Working Distance – Distance from lens to object of interest

$\boldsymbol{\alpha}$ **–** Angular acceleration of flashlamp

$\boldsymbol{\delta}$ **–** Interference between glass and polyurethane

$\boldsymbol{\nu_i}$ **–** Poisson's ratio of glass

$\boldsymbol{\nu_o}$ **–** Poisson's ratio of polyurethane

$\boldsymbol{\omega_r}$ **–** Maximum angular velocity of rotary table

1. Introduction

Lawrence Livermore National Laboratory (LLNL) is located in Livermore, California, and supports a broad spectrum of national security missions, with a primary emphasis on nuclear deterrence. The National Ignition Facility (NIF) is one of the major experimental facilities at LLNL. At this facility, high-energy physics and inertial confinement fusion research are conducted using the world's most energetic laser system, which can replicate conditions similar to a star's internal core.

The laser system begins with a seed pulse of approximately 1 *nJ* and is amplified through seven discrete gain stages. NIF employs 192 high-energy laser beams powered by 7,600 xenon filled flashlamps, to store a total of 400 *MJ* of electrical energy. This enables amplification in the final gain stage to achieve a pulse energy of approximately 10 *kJ*. Following amplification, the beams undergo three stages of frequency conversion using large potassium-dihydrogen-phosphate (KDP) and deuterated KDP crystals, shifting the wavelength from the fundamental infrared to the third-harmonic ultraviolet. Once frequency conversion is complete, the beams are focused on to a deuterium-tritium (DT) target to initiate inertial confinement fusion.

2. Problem Definition

LLNL flashlamps are made of glass or fused quartz tubes with a tungsten electrode filled with xenon gas. NIF uses over 7600 flashlamps at their facility and each flashlamp undergoes manual inspection taking approximately 2 hours per lamp, shown in Figure 1. There are over 30 kinds of categorized defects, like the ones shown in Figure 2, with some leading to catastrophic disasters if unnoticed. The need from LLNL is to improve the cycle time of the flashlamp inspection process. The goal for this project is to implement a design solution for an automated microscopic flashlamp inspection system that meets the project requirements.

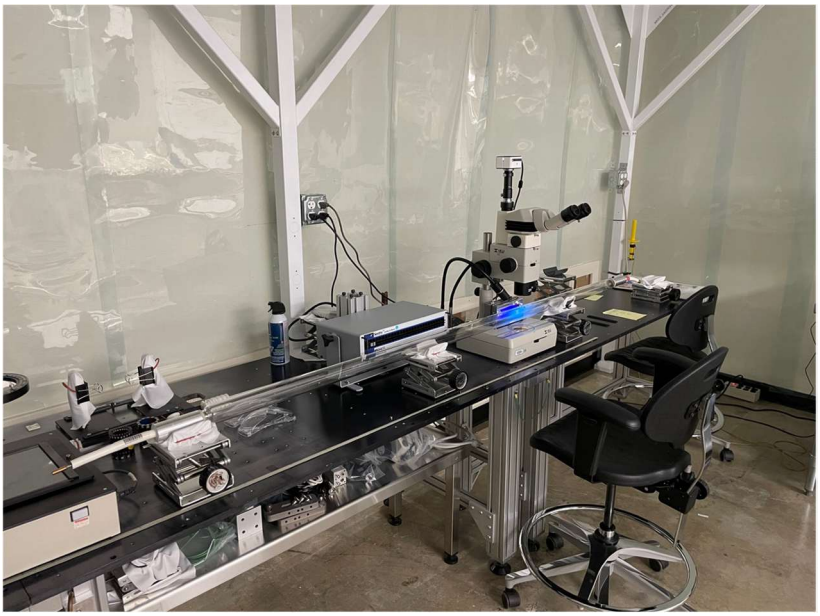


Figure 1. Current LLNL manual inspection set up.

Some requirements for the solution include lowering the cycle time to less than one hour, identifying size and location of defects, accommodating both flashlamp and tube, and finding defects that are as small as 0.3 *mm*. All project requirements are listed in Appendix A. Included in these requirements are the design specification and testing methods.

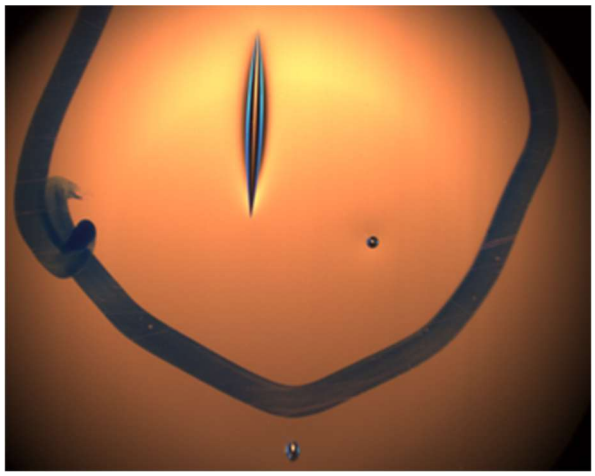


Figure 2. Defects circled by sharpie, "air line" and "bubble" defects ~ 0.3-10 *mm.*

2.1 Engineering Standards

Several required engineering standards have been identified by LLNL that ensure a safe and quality final product. Listed below are the titles of the engineering standards that need to be upheld. All standards are from LLNL Design Safety Standards: Chapter D, Section 1.1: Safety Factors, Section 1.2: Threaded Fasteners and Section 5.7: Welded Assemblies. Other standards we must adhere to are UL listed electronics and ISO 6 cleanroom classification.

3. Alternative Concepts

During the initial concept development phase, the critical subsystems were divided into two primary functional categories: flashlamp rotation and camera translation. Multiple design approaches were evaluated, drawing inspiration from established industrial mechanisms to ensure reliability, manufacturability, and precise motion control.

For the flashlamp rotation subsystem, inspiration was taken from continuous-rotation roller assemblies (e.g., hot-dog rollers), self-centering lathe chucks, and adjustable pipe-clamping mechanisms, each offering different mechanical advantages in rotational stability and fixture repeatability.

For the camera translation subsystem, design considerations were informed by belt-and-pulley drives, rack-and-pinion linear actuators, and ball-screw positioning systems, all of which are commonly employed in CNC, automated inspection equipment, and high-accuracy linear motion.

3.1 Hot Dog Roller

The hotdog roller design consists of a gear powered transmission to rotate two rollers in unison. The concept drew inspiration from hotdog warmers that rotate hotdogs. To ensure that the flashlamp does not ride up during the rotation process, a twist clamp is used to apply even pressure across the top of the flashlamp. The rollers and contacting face of the clamp would be made of rubber material to allow for safe rotation during operation. The initial sketch can be seen below in Figure 3.

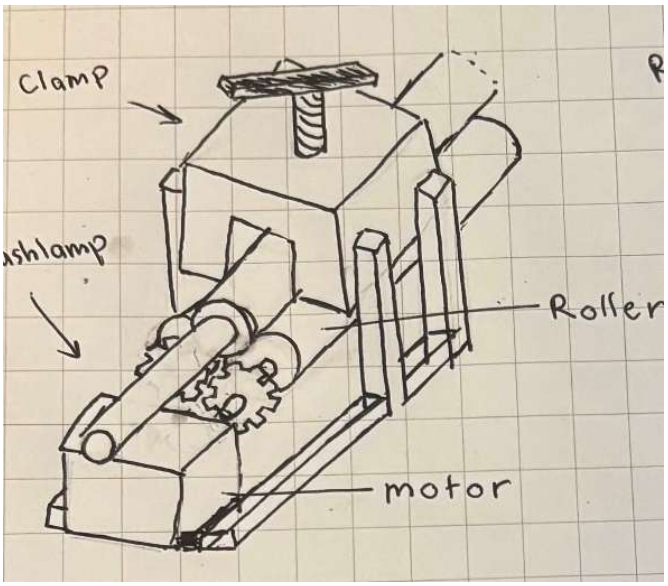


Figure 3. Hotdog roller initial sketch.

This concept provides a simple and inexpensive solution that meets many requirements for the project. However, this design introduces some noticeable constraints due to the rotational accuracy being minimal. It is unknown if the rollers will be able to maintain consistent friction while rotating the flashlamp without slipping. These uncertainties present potential risks during rotation accuracy and long-term reliability. The CAD concept model can be seen below in Figure 4.

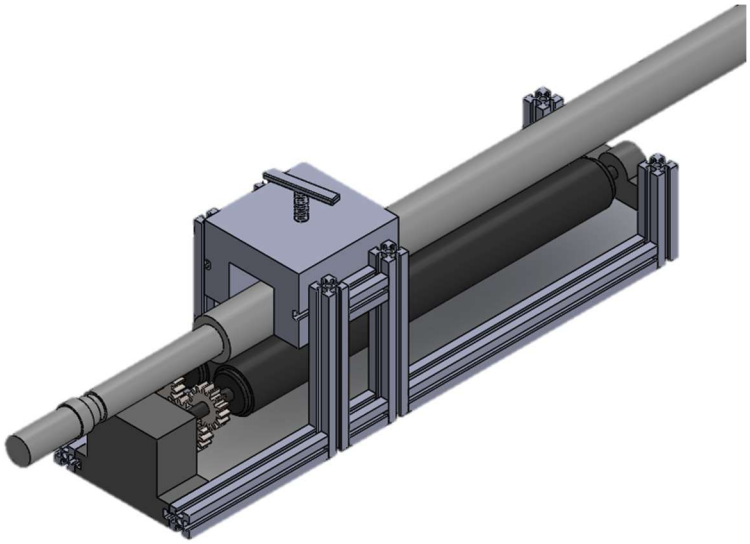


Figure 4. Hotdog roller CAD model.

3.2 Lathe Chuck

The proposed lathe-chuck solution was derived from established industrial machine lathe designs. The three jaw concept sketch presented in Figure 5 enables technicians to apply controlled, repeatable clamping forces to either the flashlamp or the tube. The jaw geometry is configurable, allowing all three jaws to be replaced or inverted to accommodate both component types while maintaining secure and non-damaging engagement.

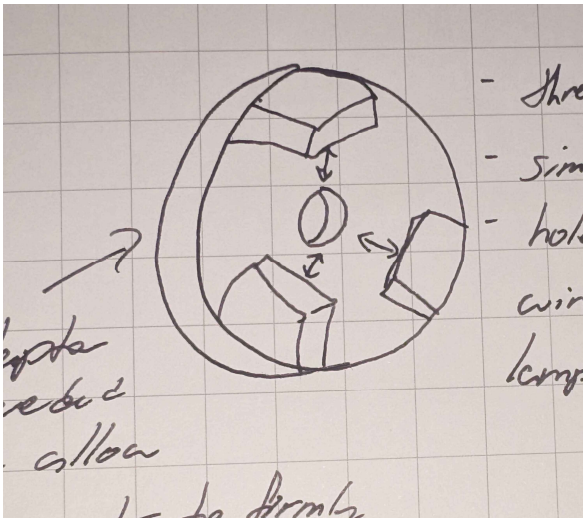


Figure 5. Lathe chuck initial sketch.

Despite the functional advantages of this approach, no commercially available lathe chuck met the required dimensional and material specifications at an acceptable cost. A custom chuck could be engineered and fabricated to satisfy these requirements, doing so would consume a large portion of the project budget, whereas the same cost could be utilized in a cheaper alternative solution. A preliminary CAD concept model illustrating the proposed configuration is provided in Figure 6**.**

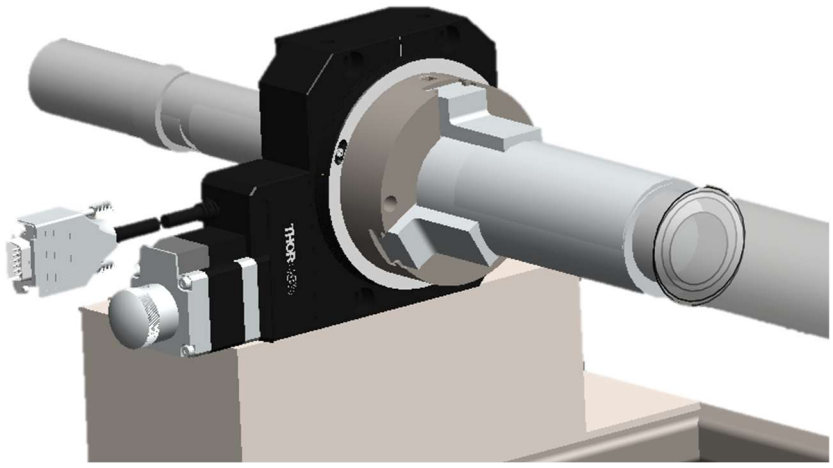


Figure 6. Lathe chuck CAD model.

3.3 Clamp

The clamp design is based on a standard 2-part pipe clamp. It consists of two semi-circular halves that hinge along one end. This design allows fine tuning of pressure based on clearance and latching pressure, while remaining simple and easy to manufacture. The clamp can be easily attached to a rotary table for precise rotation. An initial sketch can be seen in Figure 7.

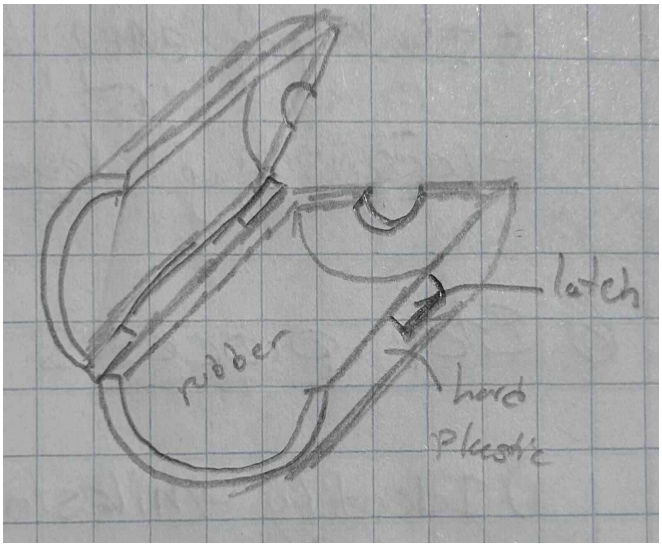


Figure 7. Clamp initial sketch.

The clamp can be 3D-printed at a low cost while holding precise dimensions. Two separate clamps will be necessary, one for holding the metallic housing on the end of the flashlamp, and one for holding the glass tube for an unfinished lamp. A layer of thin rubber or cloth will be adhered to the inside of the clamp to prevent scratches or damage to the flashlamp, while also ensuring consistent pressure across the full surface area. A CAD model mockup is shown in Figure 8.

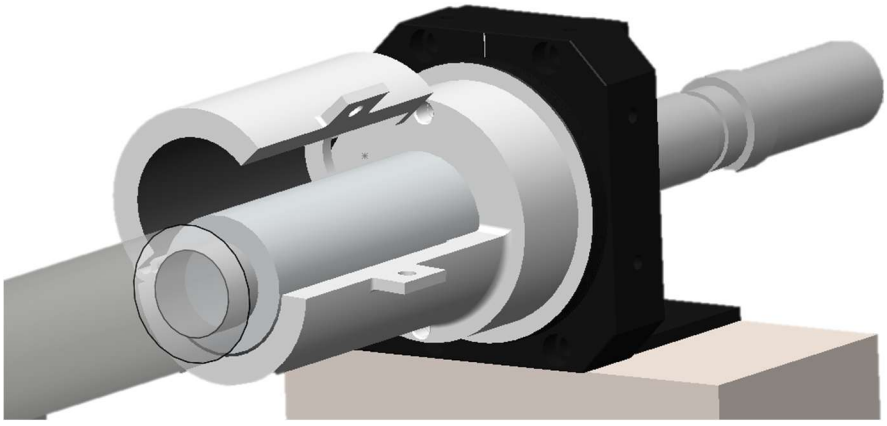


Figure 8. Clamp CAD model.

3.4 Belt and Pulley

The belt and pulley system is an effective concept for translation of the camera across a 2100 *mm* distance. It can provide smooth, precise, and low noise movement. Belts can naturally absorb small vibrations in the motor which can help steady the movement of the camera. An initial belt and pulley concept sketch can be seen below in Figure 9.

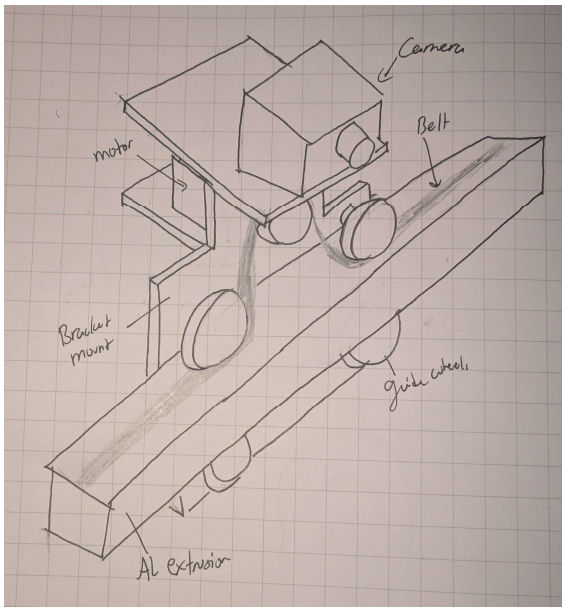


Figure 9. Belt and pulley initial sketch.

In this system, the belt-and-pulley mechanism would position the camera above the drive assembly, with a motor system located directly below. The CAD model for this concept can be seen in Figure 10.

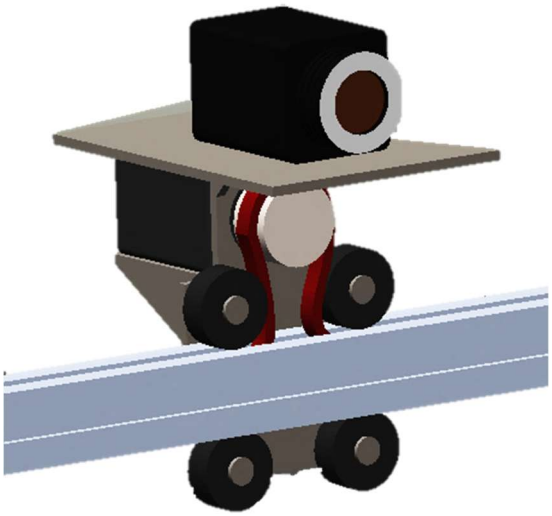


Figure 10. Belt and pulley CAD model.

The belt and pulley system presents several drawbacks. Exposure to UV light significantly reduces belt lifespan, and the tension required for proper operation increases maintenance frequency and accelerates component wear. Additionally, systems of this type exhibit poor performance at low translational speeds, resulting in decreased positional accuracy over time.

3.5 Rack and Pinion

The rack and pinion concept shown in Figures 11 and 12 provides a robust and mechanically straightforward linear translation system. Employing a helical rack-and-pinion rather than a standard spur configuration, yields reduced backlash, measured at approximately 0.06 *mm*, while enabling continuous travel of the full 2100 *mm* stroke. The helical geometry also contributes to improved smoothness of motion and an extended operational lifespan.

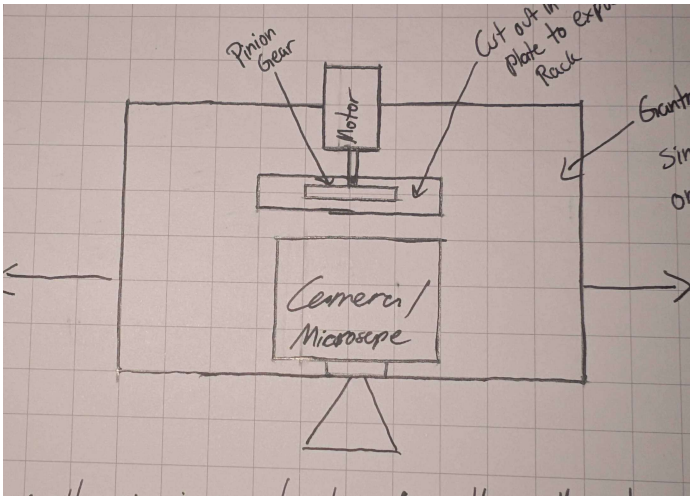


Figure 11. Rack and pinion initial sketch.

Several drawbacks were identified in the evaluation of this design. The system carries relatively high machining costs for multiple segments, with quoted prices ranging from $2,000 to $2,800. Additionally, the rack features mounting holes oriented only perpendicular to the helical threads, complicating alignment and installation. The design also includes numerous critical dimensions, if fabricated or assembled out of tolerance could increase backlash, or reduce transmission efficiency.

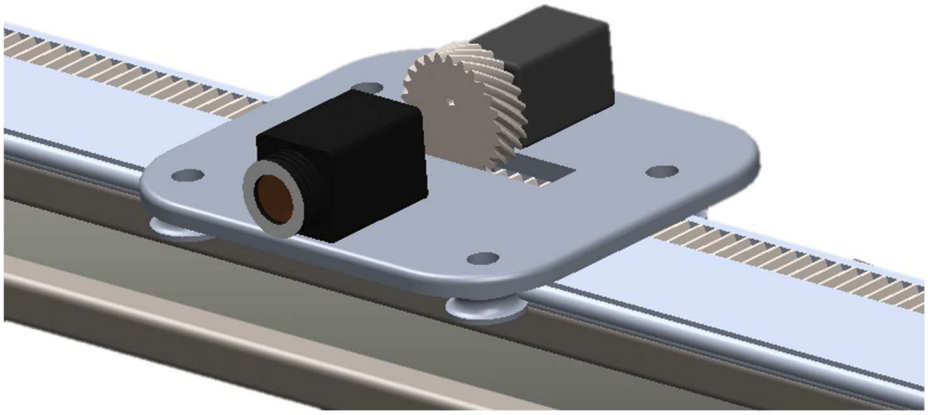


Figure 12. Rack and pinion CAD model.

3.6 Ball Screw

The proposed ball screw concept drew design inspiration from CNC machine tools and robotic actuation systems. The ball screw assembly offered several advantageous performance characteristics, including minimal backlash, (approximately 0.023 *mm*), high positional repeatability, and a dynamic load capacity of 13.2 *kN* for a screw with a 25 *mm* diameter. The system supports relatively rapid translation and demonstrates an exceptionally long calculated service life of 1.48×10^{20} *mm* of cumulative travel for the same 25*mm* ball screw resulting in 5×10^{15} flashlamps. An initial concept sketch can be seen below in Figure 13.

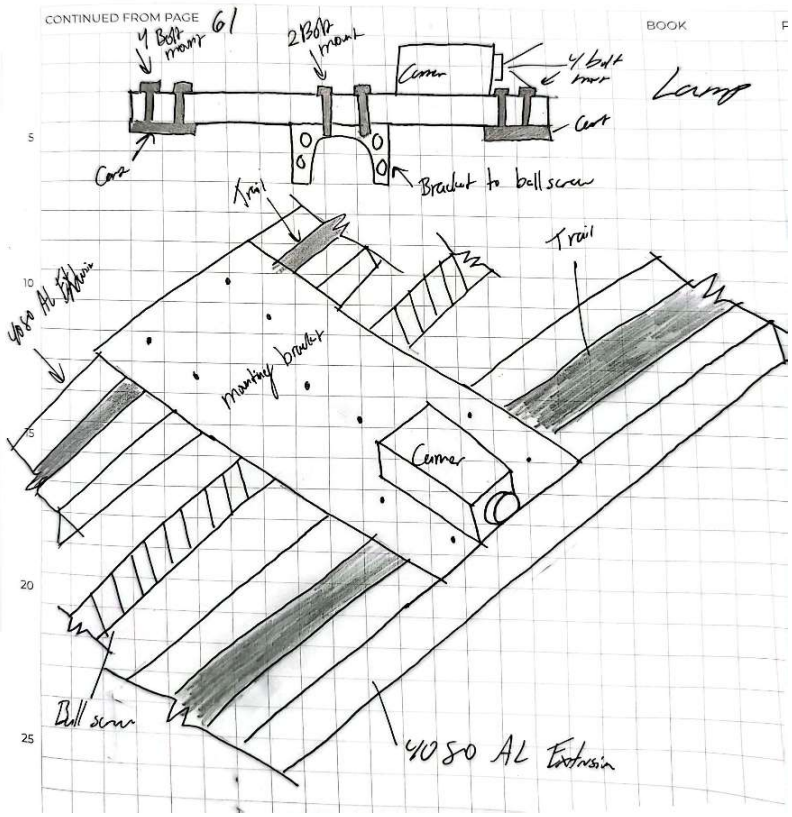


Figure 13. Concept drawing of ball screw assembly.

Despite these benefits, the design presents notable drawbacks. Ball screws of this length fall within a price range of $2,500 to $3,500, representing a significant portion of the overall budget.

Furthermore, extended lead times for procurement may introduce substantial time delays, creating bottlenecks in the assembly process and reducing available time for integration and testing. A preliminary CAD concept model illustrating the proposed configuration is provided in Figure 14**.**



Figure 14. Ball screw carriage CAD model.

3.7 Imaging

Several lensing options were evaluated, the main delineation made was fixed focal length, variable focal length, and telecentric. Fixed focal length lenses are used in machine vision applications which take in a cone of light and focus it onto a sensor. They are used in a wide variety of applications due to loose requirements for working distances, often relying more on calibration than being engineered for a specific goal. Major drawbacks of fixed focal length lenses are that there can be parallax issues; A closer object may appear larger than a further one, even if the further one is greater in size. This issue largely rules out the use of fixed focal length, as measuring the size of defects accurately is paramount to the project.

Variable focal length lenses have the advantage of altering their focal length, effectively being able to zoom in or out either manually or electrically. It was determined that the variable magnification range does not benefit the project nor justify the required cost.

A telecentric lens takes in parallel rays and focuses them onto a sensor. Since they accept parallel rays, the parallax error inherent to fixed focal length lenses doesn't occur. Due to the decreased error, telecentric lenses are commonly used in precision metrology applications. Figure 15 demonstrates the difference between fixed focal length and telecentric, presenting the benefit of telecentric lenses.

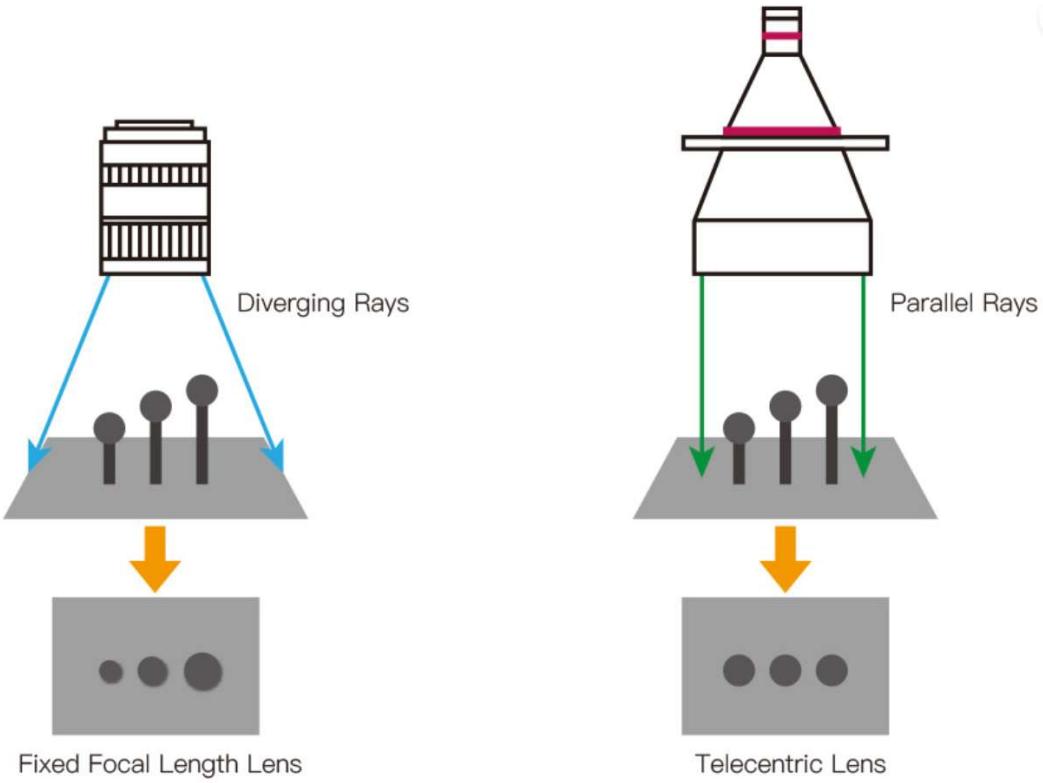


Figure 15. Fixed focal length versus telecentric lens.

For a fixed focal length, features incur error in dimensional accuracy when they are different distances. For this application, the error can be significant due to the curved surface of the flashlamp, creating calculated error in object size representation. The major drawbacks of

telecentric lenses consist of cost and low tolerance working distances. These drawbacks are manageable, since the lens is within budget and the rest of the system can be designed around the working distance of the specific lens.

4. Designed Solution

The final designed solution utilizes aluminum T-slotted extrusion base. The T-slot extrusion is dimensionally consistent, easy to procure, and provides quick assembly. The flashlamp is supported on both ends with polyurethane coated rollers, safely supporting the lamp while rotating. The rotation is induced by a custom 3D printed clamp and a Thor Labs rotary stage, fixed to the frame with a custom aluminum bracket. The components are mounted to the frame using standard extrusion gussets and T-nuts. Two rails and sliders connected by a carriage base plate support a telecentric imaging system for smooth translation across the length of the flashlamp. This translation is controlled using a ball screw and stepper motor. A custom fixture connects the bottom of the carriage plate to the ball screw. LabVIEW software controls all motion, data collection, and imaging simultaneously. A live camera feed is fed to the existing workstation at LLNL. A CAD model can be seen below in Figure 16.

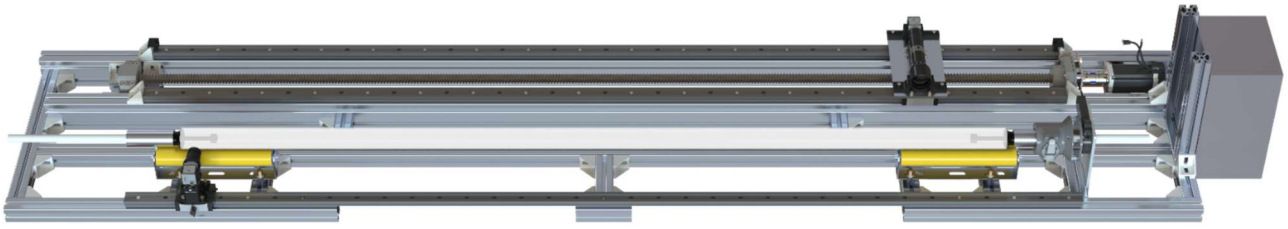


Figure 16. Designed solution CAD model.

Working drawings for all assemblies, subassemblies, and custom parts can be found in Appendix B. Conceptual design selections were made in accordance with the Pugh Matrices provided in Appendix C

4.1 Rotational Solution

4.1.1 Purchased Parts

Polyurethane Support Rollers

The Ultimation 150R-10BR-PU Polyurethane Rollers were chosen to support the flashlamp. Without the rollers the flashlamp would become a cantilever beam, which is unsafe and may break the flashlamp. The Polyurethane coating on the outside will prevent damage to the flashlamp during operation while maintaining friction to prevent slippage. The rollers have bearings on either side to ensure that smooth rotation occurs along the 10 *in* roller. The 150R-10BR-PU satisfies the ISO 6 clean room requirement. They can be seen below in Figure 17.



Figure 17. Ultimation 150R-10BR-PU.

Precision Rotary Stage and Controller

A ThorLabs PRM1Z8 rotation stage shown below in Figure 18 includes its own integrated motor and gearbox, allowing precise and stable rotational movement to 0.0005°. This built-in motor system enables smooth and accurate angular positioning without the need for an

external drive mechanism. The PRM1Z8 is controlled directly by a ThorLabs KDC101 controller/driver, which sends the necessary command signals to regulate rotation speed, direction, and position. The ThorLabs Controller/Driver is shown in Figure 19 below. By combining an integrated motor with teensy-based control board, the rotation stage offers reliable, repeatable motion essential for the system's performance requirements.

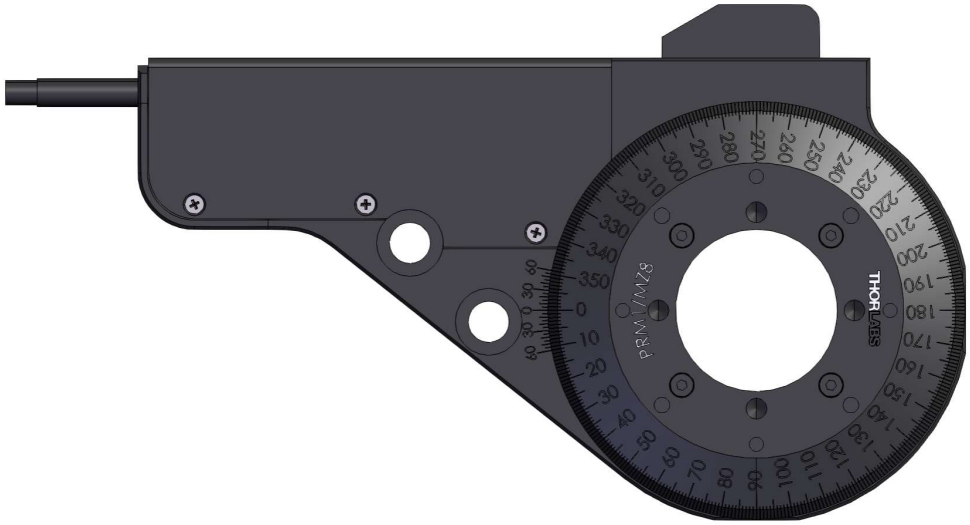


Figure 18. ThorLabs PRM1Z8 motorized rotation mount.



Figure 19. ThorLabs KDC101 controller/driver.

Rubber Standoffs

Four 9232K653 natural rubber vibration-damping sandwich mounts are used to isolate vibration and provide compliance within the custom clamp assembly. The standoffs have a durometer hardness of 55A and separate the rotary stage mounting plate and the clamp assembly. They are shown below in Figure 20.



Figure 20. 9232K653 Natural rubber vibration-damping sandwich mount.

4.1.2 Custom Parts

The following sections are about custom components which are designed for the rotation subsystem of this solution. All custom parts will be manufactured in house at CSUC using CNC milling and 3D printing.

Rotary Mount

The rotary frame mount is made of 0.250 *in* 6061-T6 aluminum, and mounts the ThorLabs rotary stage to the aluminum extrusion frame providing stability and minimal deflection. The rotary mount is pictured below in Figure 21.



Figure 21. Rotary table frame mount.

Clamp and Clamp Fixture Plate

The clamp being used for the main clamping force during rotation will be 3D printed out of polycarbonate (PC). PC is ISO 6 clean room compatible, UV resistant, and durable. There are two separate clamps, one designed to hold standard flashlamps and another to hold tubes. On the inside of the clamp there is a layer of polyurethane rubber to prevent any damage and apply even clamping pressure. The polyurethane rubber is adhered to the surface of the clamp with a specialized J-B Weld plastic bonder polyurethane epoxy rated to 3770 *psi*. The flashlamp clamp utilizes 0.125 *in* durometer 40 OO polyurethane rubber while the tube clamp uses 0.1875 *in* durometer 30 OO rubber. The clamps use a spring-loaded stainless-steel draw latch to lock the clamp closed during operation while remaining easy to actuate. The draw latch is bolted to the clamp using threaded heat inserts to prevent damage to the clamp and increase the life cycle. The clamp bolts to the rubber vibration-damping standoffs for compliance when loading and operating the system. The standoffs bolt to a PC fixture plate which bolt directly to the ThorLabs

rotary stage. Both clamps and the fixture plate are shown below in Figure 22, 23, 24 and 25.

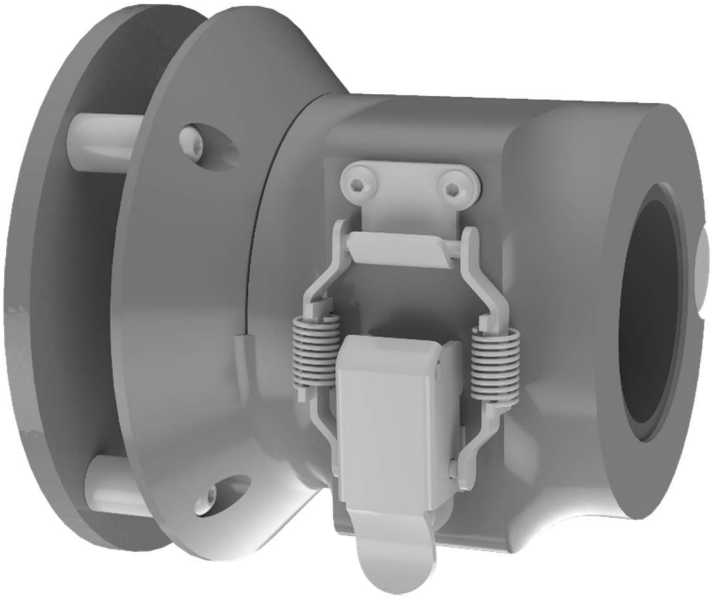


Figure 22. Flashlamp clamp assembly.

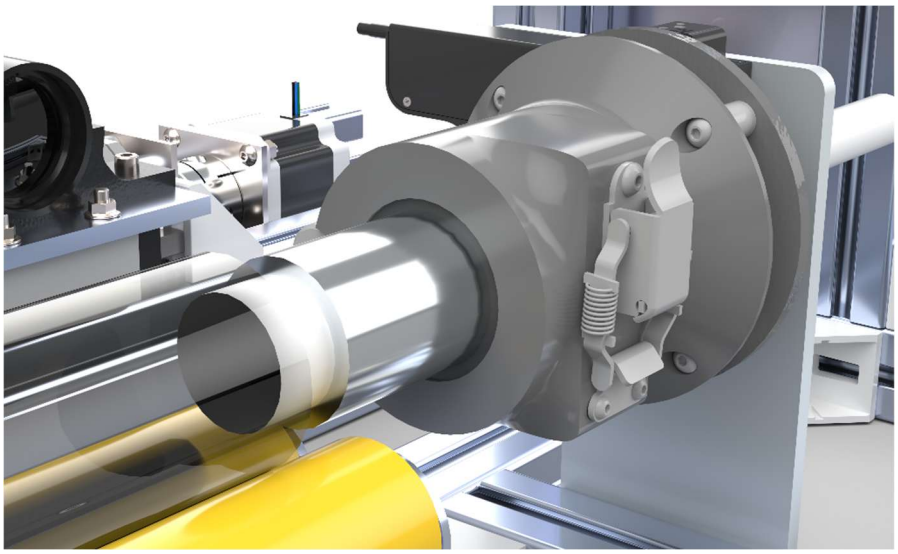


Figure 23. Flashlamp clamp assembly on frame.

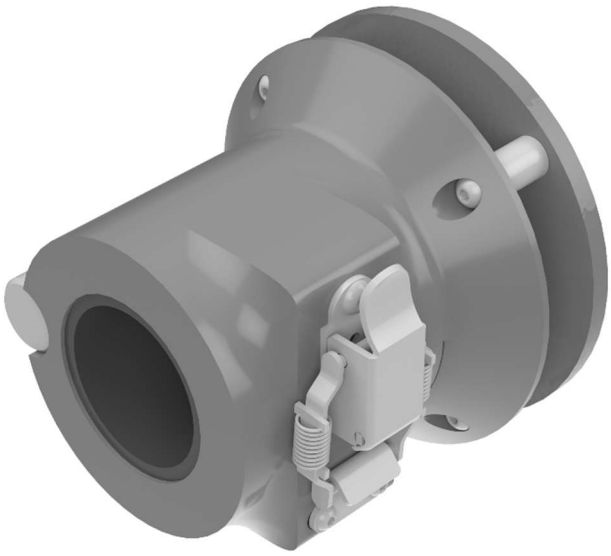


Figure 24. 3D printed clamp made of PC.



Figure 25. PC clamp fixture plate.

4.1.3 System Solution

Shown below in Figure 26 is a full rendering of the rotational subsystem solution.

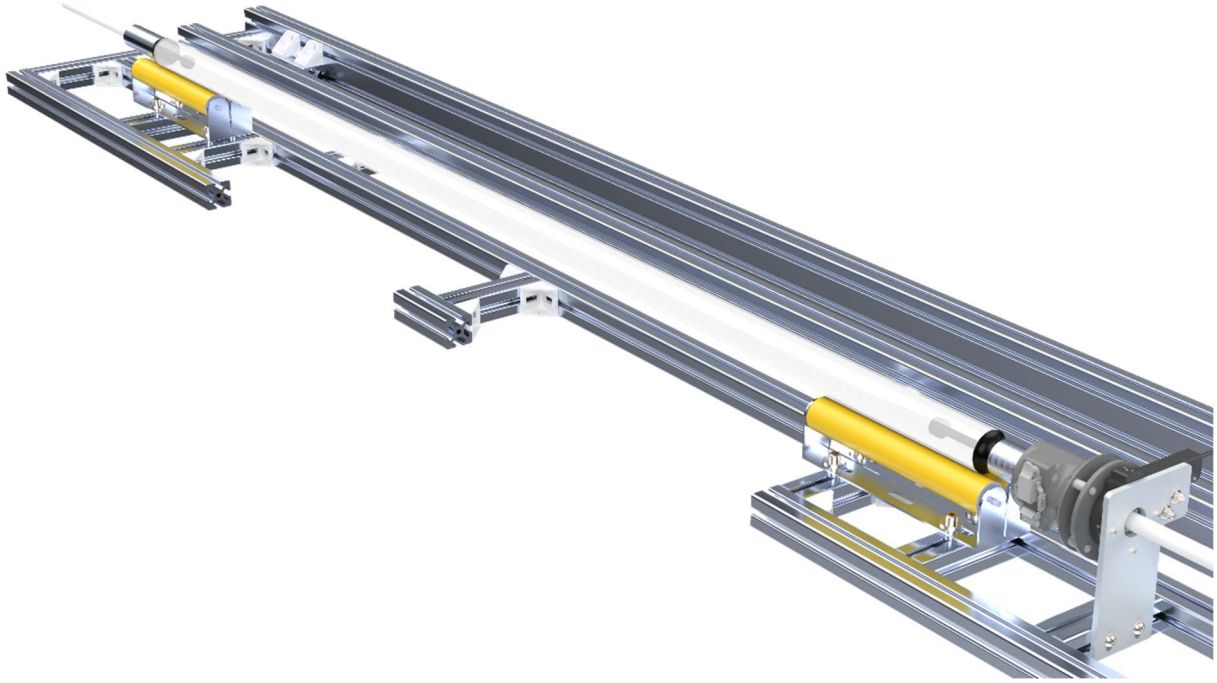


Figure 26. Rotation subsystem assembly on frame.

4.2 Motion Controls

4.2.1 Hardware

Teensy 4.1 serves as the main controller for the translation motor control system which can be seen below in Figure 27. Its high processing speed, compact footprint, and extensive digital I/O capability makes it well suited for precise motion control applications. The Teensy 4.1 can also communicate with LabVIEW through serial commands which allows motor commands, position parameters, or system inputs to be taken from a LabVIEW user interface.



Figure 27. Teensy 4.1.

The AMT222B-V absolute encoder was incorporated into the system to provide reliable angular positional feedback of the motion control assembly. As an absolute encoder, it tracks the shaft's position directly, rather than tracking only relative changes in rotation. This allows the controller to determine the encoder position immediately after power-up without requiring a reference or homing sequence. The AMT222B-V is used to monitor the rotational position of the motor shaft, providing feedback that supports accurate camera translation across the linear travel range. The supplied position data improves system repeatability, verifies commanded motion, and contributes to the overall accuracy and reliability of the camera positioning system. The AMT222B-V can be seen below in Figure 28.



Figure 28. AMT222B-V absolute encoder.

The motor selected for the translation of the camera is a StepperOnline 23HS45-4204D-E1000 NEMA 23 closed-loop stepper motor, 3.0 *N-m*, shown below in Figure 29. This motor provides sufficient torque capacity and closed-loop feedback capability, which improves positional accuracy and helps prevent missed steps during operation. The stepper motor has a 1.8° step angle and is equipped with an integrated encoder providing 4000 counts per revolution. The 3.0 *N-m* holding torque and closed-loop encoder feedback make it suitable for maintaining smooth, accurate, and repeatable camera motion along the linear translation system.



Figure 29. Nema 23 closed loop stepper motor 3.0 *N-m*.

The CL57T closed-loop stepper motor driver is used to control the stepper motor within the camera translation system and is shown below in Figure 30. The driver receives step and direction signals from the controller and converts them into appropriate phase currents required

to drive the motor accurately and efficiently. The CL57T uses encoder feedback to monitor the motor's actual position and correct for positional error during motion. The stepper driver provides necessary interface between the Teensy 4.1 controller and the closed-loop stepper motor, supporting smooth, repeatable, and precise camera movement along the translation axis.



Figure 30. CL57T closed loop stepper motor driver.

4.2.2 Software

The LabVIEW front panel below in Figure 31 provides user controls and indicators needed to run and monitor the system. The **START CYCLE** button begins with the programmed motion sequence, while the **STOP** button immediately stops the test. The Live View area is reserved for displaying camera feedback while the program is running. On the right side of the

panel, the Status indicator shows the current operating state of the program. The “Position [*cm*]” indicates the current position of the camera along the length of the ball screw, while the “Pass # out of 15” indicates how many times the camera has translated across the lamp. The “Send Home” button commands the system to return to the home position. The manual movement for the ThorLabs Rotation stage can be seen below in Figure 32, provided via Kinesis drivers. Together these controls allow the operator to start the cycle, manually jog the stage, monitor progress, and stop the system when needed. LabVIEW can read and write serial commands to the Teensy 4.1.

The motion sequence is programmed in the Arduino IDE program and is executed using the Teensy 4.1 microcontroller. When the **START CYCL**E button is pressed in LabVIEW, the motor initiates a homing routine to establish the camera translation system’s reference position using the two limit switches. Once homing is complete, the motor translates the camera to one end of the travel range and pauses while the ThorLabs rotation stage rotates 24°. The camera then translates to the opposite end and pauses again while the rotation stage performs another 24° rotation. This back-and-forth translation and rotation sequence is repeated seven times. After the cycle is complete, the camera returns to the home end of the system and remains idle until the next command is received from LabVIEW. The “Jog Commands [*cm*]” and “Move [*cm*]” controls allow the user to manually control the motor. All movements are executed using a trapezoidal motion profile to provide controlled acceleration and constant-velocity travel.

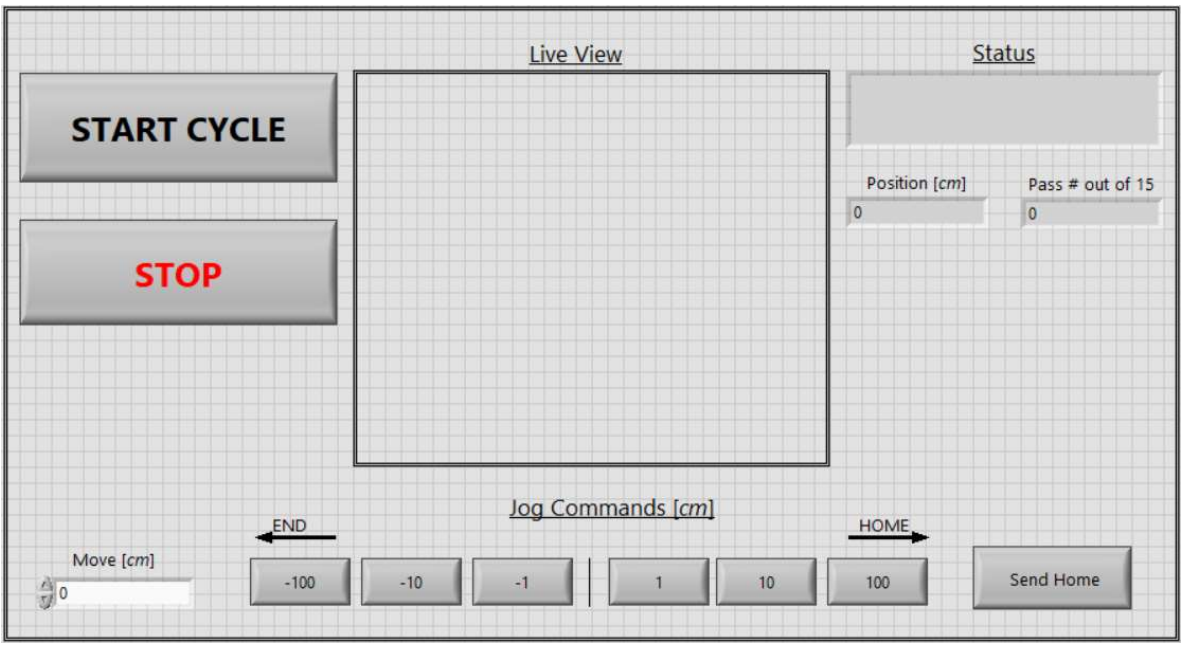


Figure 31. LabVIEW user interface.



Figure 32. Manual movement controls for ThorLabs rotation stage in LabVIEW.

4.2.3 Custom Parts

The custom PCB, which includes the Teensy 4.1 pin connections and the 30 V to 5 V converter, is shown in Figure 33. The Teensy and supporting components were soldered to a prototype PCB to improve wiring organization and reduce the risk of loose or disconnected wires during operation. This approach provided a more reliable and compact electrical interface for the control system. The corresponding PCB wiring diagram is provided in Appendix D.

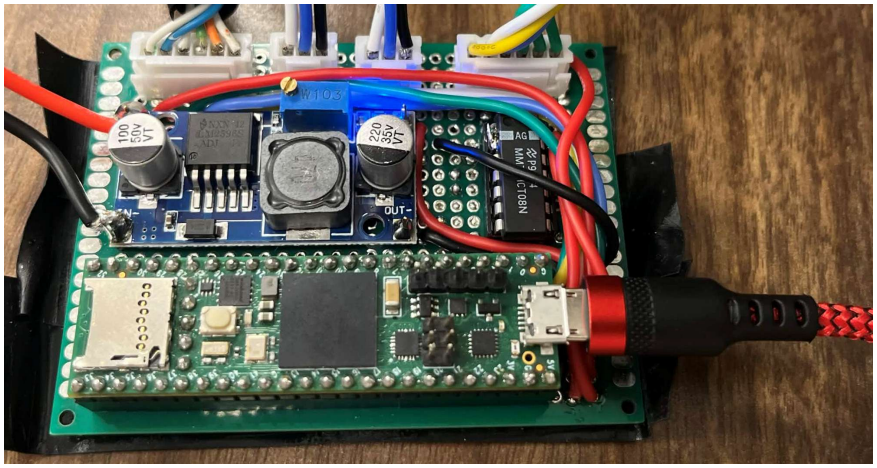


Figure 33. Custom PCB assembly.

The custom 3D printed PC parts shown below in Figures 34 and 35 were designed to support the mechanical integration of the positioning and feedback components within the system. The limit switch mount utilizes an interference fit to prevent misalignment in position upon contact. The encoder mount was designed as an adapter to properly align the encoder with the stepper motor, while preventing it from moving with the shaft.

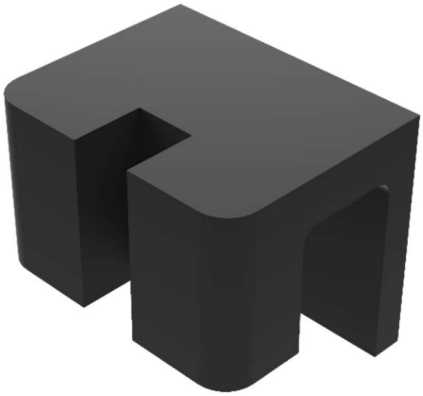


Figure 34. Limit switch mount.



Figure 35. Encoder mount.

4.3 Optics

4.3.1 Hardware

Primary Camera and Lens

The PlatinumTL Telecentric Lens from Edmund Optics was chosen for its field of view and camera compatibility. The working distance is 173 *mm* with a depth of field of $\pm$2.1 *mm*. The concavity of the flashlamp creates a changing working distance across an image. Calculations verify that this difference is within the depth of field. The PlatinumTL is shown below in Figure 36.



Figure 36. 0.5X, 2/3" C-Mount PlatinumTL telecentric lens.

The Basler ace2 a2A4200-40ucBAS color USB3 camera was chosen due to its high resolution, framerate, and USB capabilities. This camera, shown in Figure 37, is also chosen for its compatibility with the selected PlatinumTL lens. The C-Mount is a standard thread pattern for cameras and lenses to allow easy compatibility and interchangeability between other C-Mount lenses.



Figure 37. Basler ace2 a2A4200-40ucBAS color USB3 camera.

Secondary Camera and Lens

An additional camera is required to inspect the glass-to-metal junction in the end cap of the flashlamp. The optical requirements for imaging the junction are not as stringent as that of the primary camera. Without these strict requirements, a fixed focal length lens is sufficient. The secondary camera motion will be controlled manually, at the request of the sponsor. The 35mm, f/1.85 sealed UCw series fixed focal length lens meets design requirements. The camera used is a Basler ace acA2500-14uc color USB 3.0 camera, with less resolution and lower framerate than the primary. The lens is shown below in Figure 38, while the camera is represented by Figure 37, since both cameras have the same form factor.



Figure 38. 35mm, f/1.85 sealed UCw series fixed focal length lens.

Manual Rotation Stage

A MSRP01/M Ø1.40" Manual Mini-Series Rotation Stage is used to rotate the secondary camera to allow inspection of both end caps on the flashlamp. It has a locking set screw to hold the stage at specific angles. The stage is shown below in Figure 39.



Figure 39. A MSRP01/M Ø1.40" manual mini-series rotation stage.

4.3.2 Image Processing

All code for the project is run on LabVIEW. LabVIEW supports the use of Python through the Python node, allowing a user to integrate their Python code directly into a LabVIEW VI. The Python libraries used in this solution are OpenCV, Ultralytics YOLOv8, and PyYAML. OpenCV and YOLOv8 are used in tandem. OpenCV can size and identify every defect type, with

its primary issue being classification. YOLOv8 is better at classification, but requires around 100 unique images per defect type, which are currently unavailable. To remedy this, the LabVIEW front panel shows any defect which YOLOv8 is unconfident in for an operator to manually select, shown in Figure 40. YOLOv8 uses these selections to become better at categorizing defects, which is known as Active Learning. While there is currently a limited number of categories, the infrastructure is embedded to create more categories for an operator to choose from. Finally, PyYAML creates a document which summarizes the amount and severity of defects. A user defined critical dimension per category allows the program to flag certain defects as critical. Finally, in LabVIEW, all images are routed into folders based on their location (each angle has its own folder), then processed images are routed into their own subfolder. All collected data exists under a master folder designated by serial number of the flashlamp. Since the full image of the lamp consumes 18 *GB* of storage, the option to delete the bulk of these images is also given to the operator, only keeping processed images.

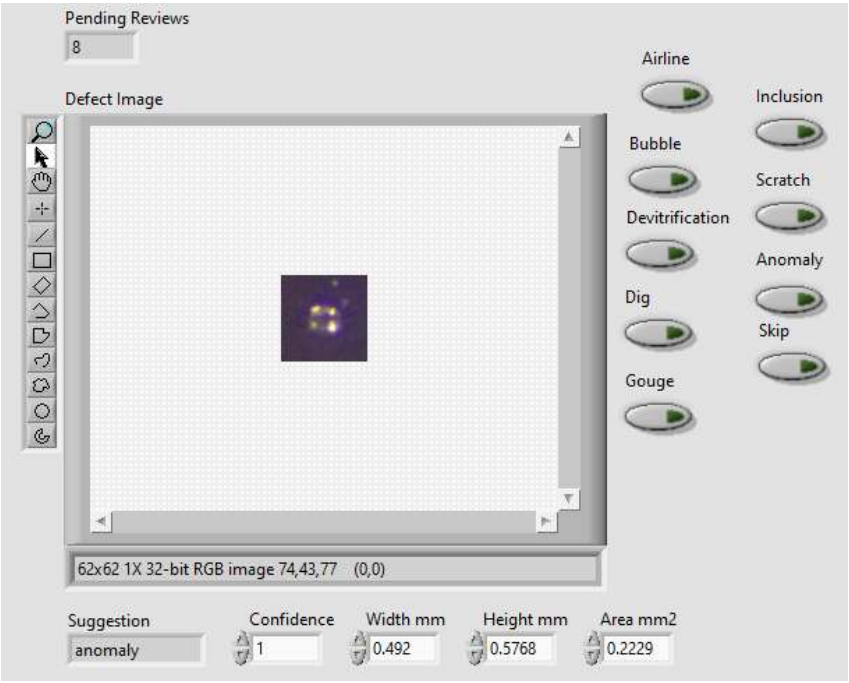


Figure 40. Manual defect categorization GUI.

4.3.3 Custom Parts

Primary Camera Mounts

The camera and lens mounts are 3D printed out of PC and utilize a press fit with the camera and lens. The primary camera and lens mounts will be fixed to the carriage and are pictured below in Figure 41 and 42.

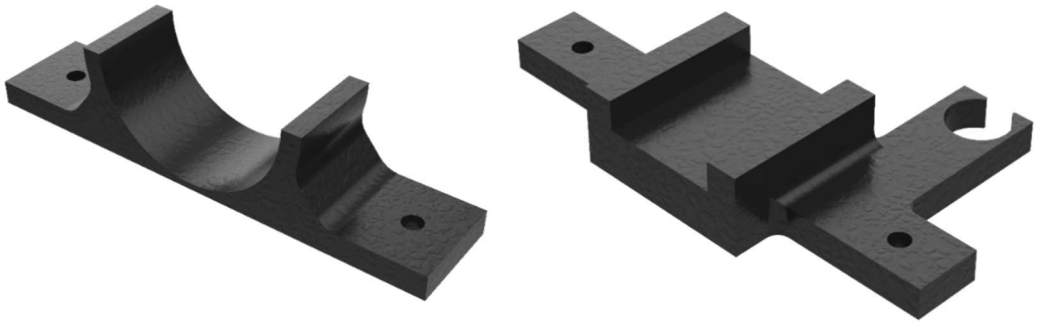


Figure 41. Camera and lens mount.

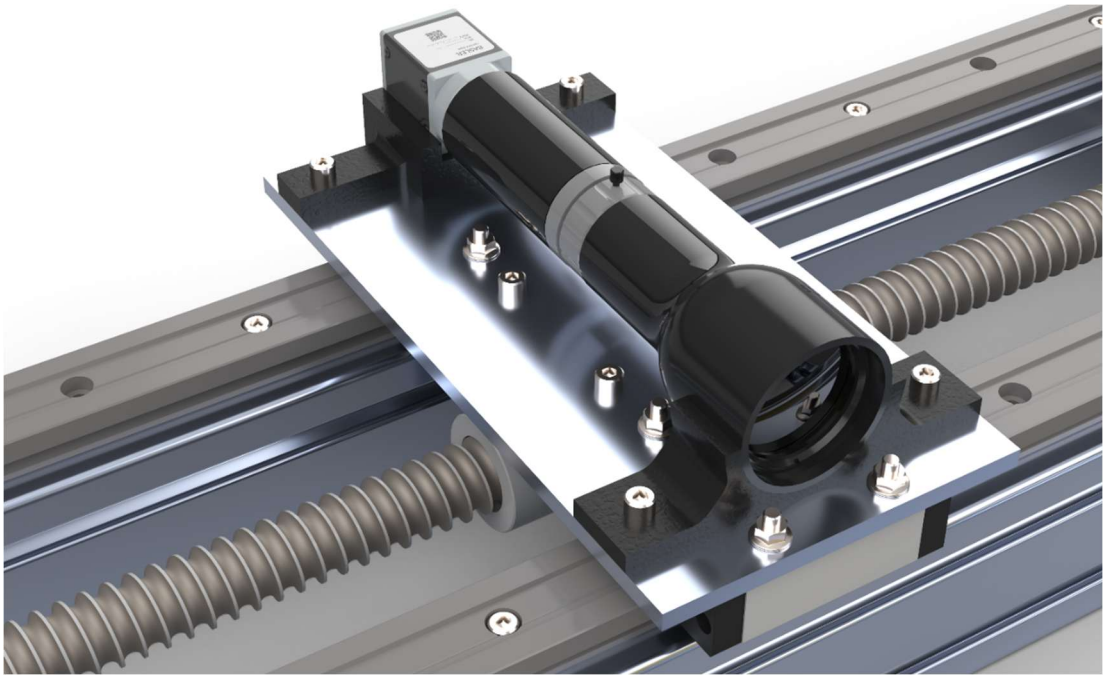


Figure 42. Primary camera assembly in position.

Secondary Camera Mount

The secondary camera and lens mount is 3D printed out of PC and utilize a press fit with the camera and lens. The secondary camera and lens mount is pictured in Figure 43.

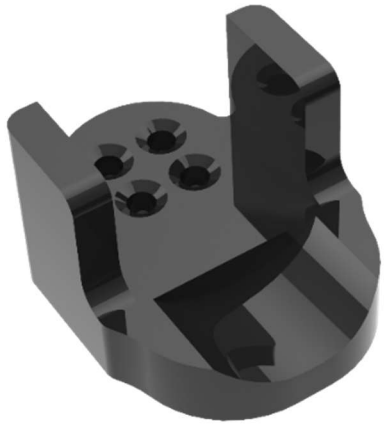


Figure 43. Secondary camera mount.

Manual Rotation Stage Mount

The MSRP01/M Ø1.40" Manual Mini-Series Rotation Stage is bolted to the manual rotation stage mount shown in Figure 44. The mount is 3D printed PC and provides the height required for the secondary camera to properly view the end caps. The secondary camera assembly is shown in Figure 45.

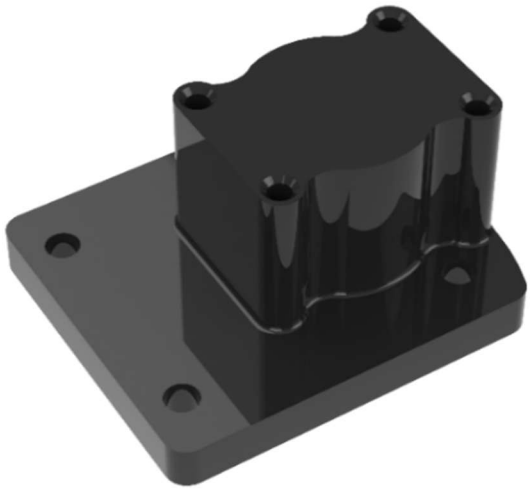


Figure 44. Manual rotation stage mount.

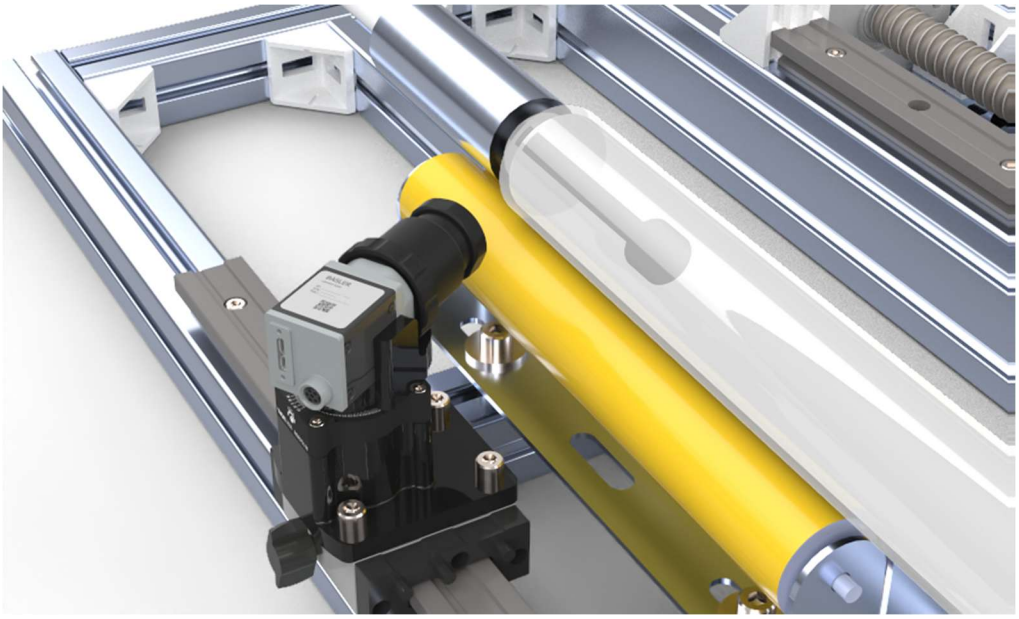


Figure 45. Secondary camera assembly in position.

4.4 Translational Solution

4.4.1 Purchased Parts

Ball Screw

The Thomson Linear KGS-2510-P5-RH ball screw has a clearance preload to ensure backlash and positional repeatability capable of satisfying system requirements. The ball screw has a dynamic load capacity of 13.2 *kN*, which is capable of translating the carriage and camera system. This system has an expected service life of $1.48 * 10^{20}$ *mm*, or approximately $5 * 10^{15}$ flashlamps. Figure 46 below is a shortened CAD model of the ball screw system used in the designed solution.

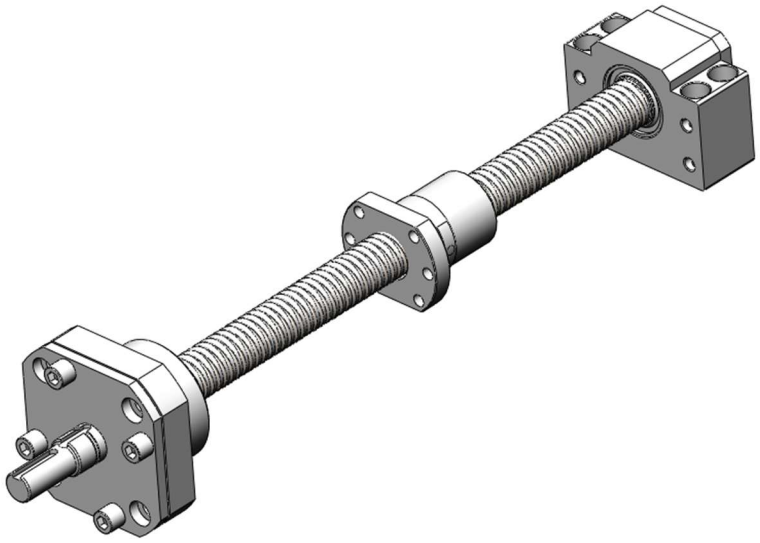


Figure 46. Thomson linear KGS-2510-P5-RH.

Rail and Carriages

Two Drylin® T Standard Guide Rail (TS-01-20) and T Standard Carriage (TW-01-20) form the base of the carriage assembly. Each rail has a slider which can freely move the full 2100*mm*. The sliders are fixed to the carriage plate, which is translated by the ball screw. Both the rails and carriages satisfy ISO 6 cleanroom requirements. Both the rail and carriage are shown below in Figure 47.

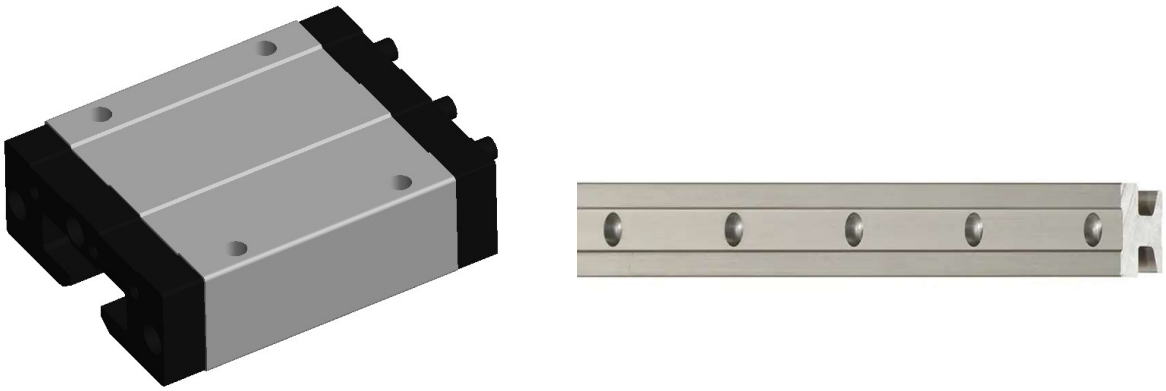


Figure 47. Drylin T standard guide rail (TS-01-20) and carriage (TW-01-20).

The front translation assembly for the secondary camera system is a manual locking Drylin T Standard Carriage (TW-01-20-HKA) and slides along another 2100*mm* Drylin® T Standard Guide Rail. This system remains manually operated at the request of LLNL. The carriage can be seen below in Figure 48.

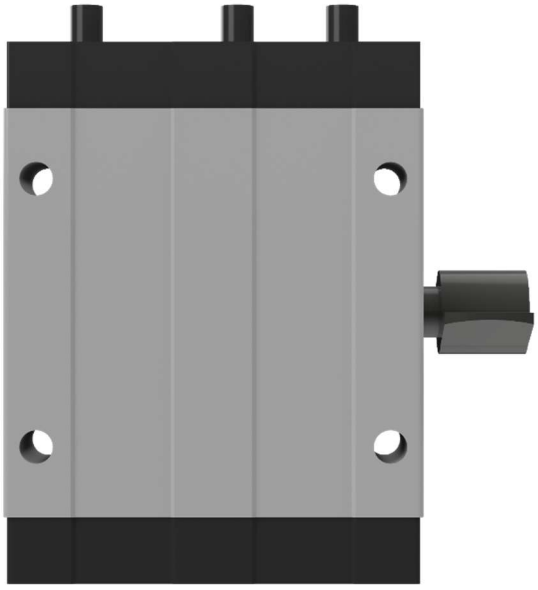


Figure 48. Manual Drylin carriage.

4.4.2 Custom Parts

Carriage Fixture Plate and Flange Mount

The carriage connecter baseplate and flange mount is manufactured from 6061-T6 aluminum, laser cut and CNC milled in house at CSUC. The purpose of the plate is to provide a mounting platform to fix the camera to the carriages and the flange mount, forming the carriage assembly. The flange mount will connect the carriage plate to the ball screw. The carriage plate and flange mount are both pictured below in Figure 49 and 50, respectively.

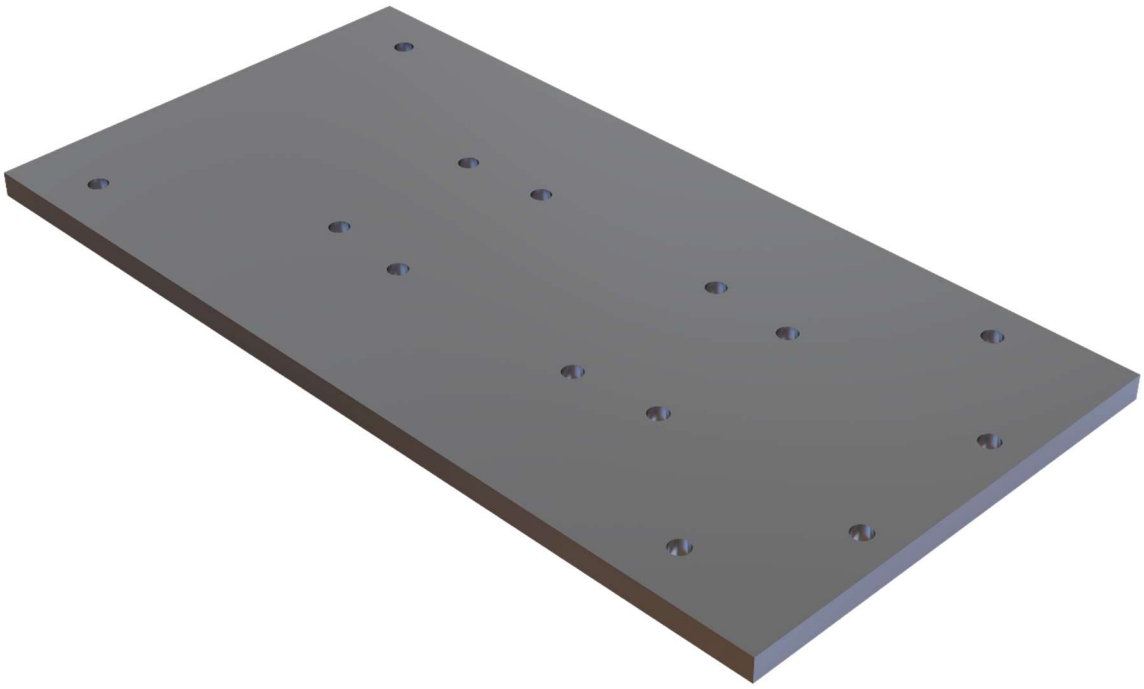


Figure 49. Carriage connector base plate.

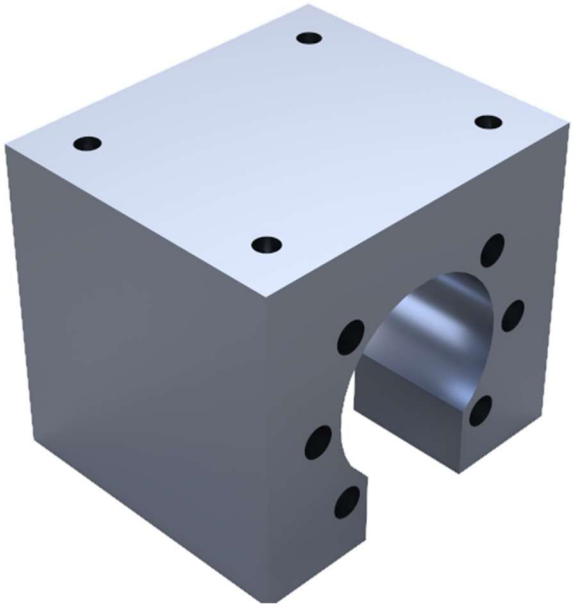


Figure 50. Flange mount.

Frame Mounting Brackets

To securely mount the Thomson linear ball screw and Nema motor, custom mounting brackets were designed and fabricated at CSUC. The mounting brackets are made from 0.125 *in* 316 Stainless steel. The material choice was dictated by the ISO 6 cleanroom requirement; 316 stainless steel does not rust or release particulate after welding. The two brackets can be seen below.

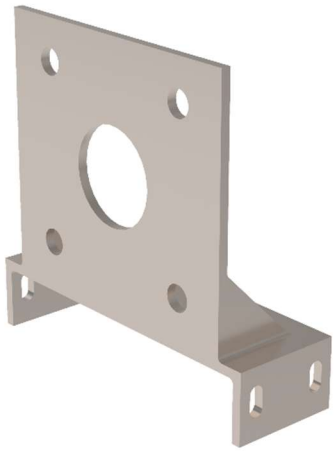


Figure 51. Ball screw mounting bracket.



Figure 52. Nema motor mounting bracket.

4.4.3 System Solution

Shown below in Figure 53 is a full rendering of the translation subsystem solution.

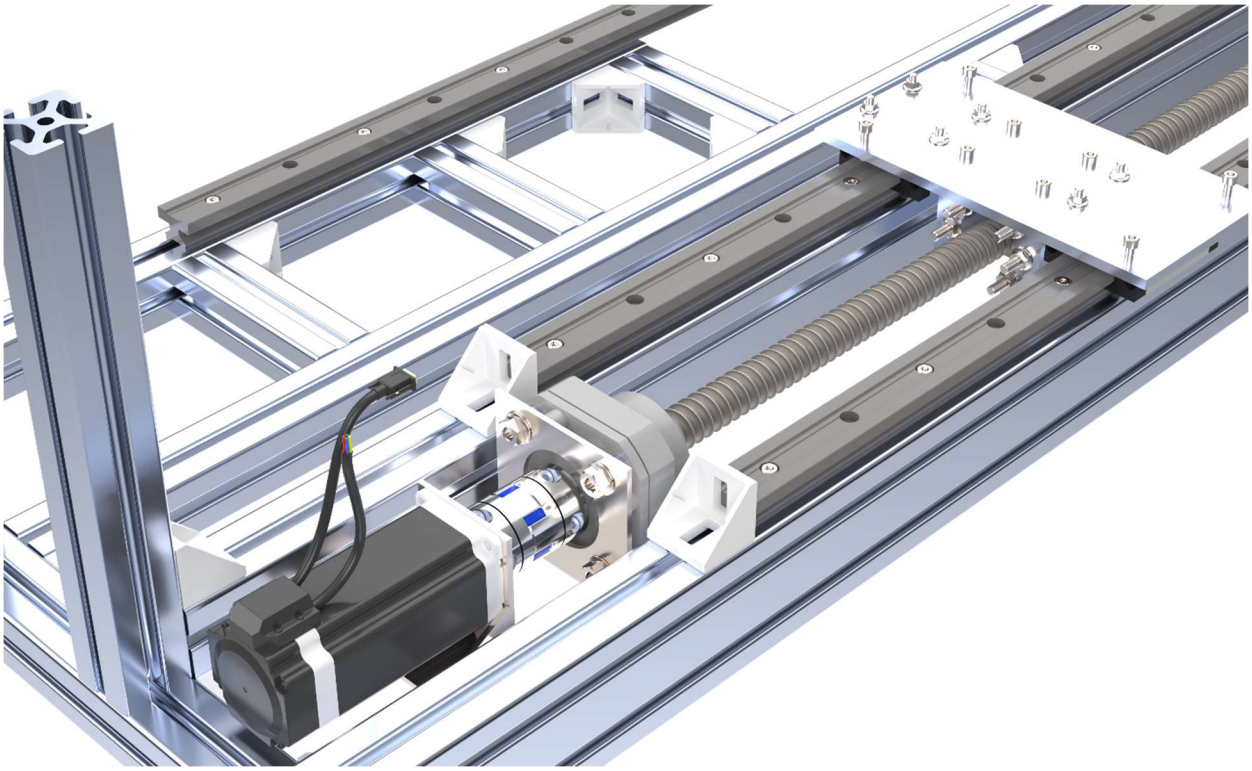


Figure 53. Linear translation assembly.

4.5 Frame

T-Slotted Aluminum Extrusion Frame

The frame is comprised of 40 series t-slotted aluminum utilizing both 4040 and 4080 extrusions. The primary camera rail system utilizes 4080 and the rest of the frame utilizes 4040. The choice of 4080 for the primary camera rail system was to provide sufficient initial height to align with the flashlamp. A CAD model of the frame is shown in Figure 54.

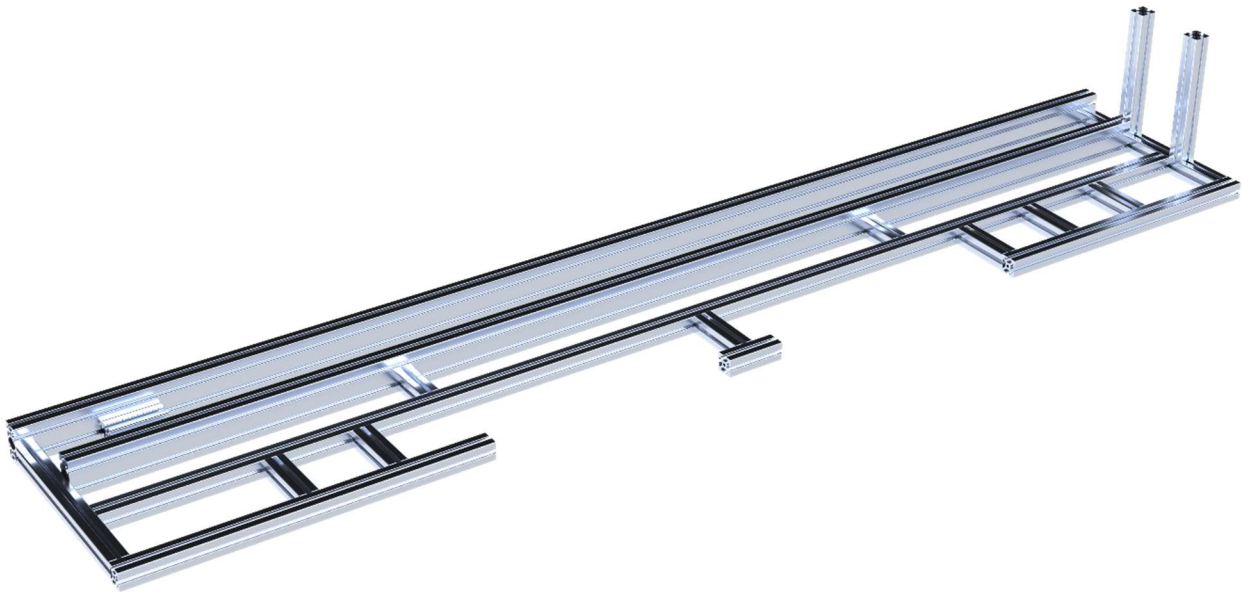


Figure 54. Aluminum extrusion frame.

4.6 Fasteners

Fasteners are standard 304 Stainless Steel metric fasteners which are in accordance with project requirements and standards. All fasteners are for locational fixtures and will not undergo significant loading.

5 Engineering Analysis

To ensure that the solution meets requirements, several engineering analyses were performed. The pressure exerted by the clamp on the tube, an estimate of the cycle time, and a defect pixel count of the lensing system are calculated.

5.1 Clamping Pressure and Torque Transmission

To ensure safety of the flashlamps during rotation, the clamping pressure applied by the clamp was calculated. A model was created of the clamp simplifying the pressure applied to a standard interference fit between the flashlamp and the layer of polyurethane on the inside of the clamp, shown below in Figure 55. Values used for calculation are listed in Table 1.

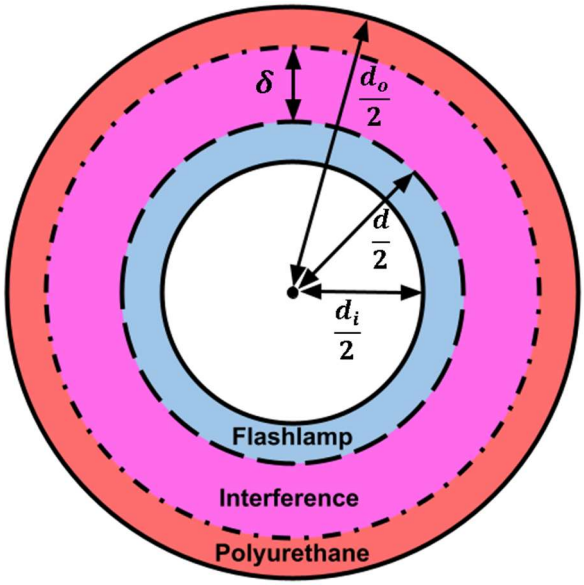


Figure 55. Standard interference fit between flashlamp and polyurethane inside the clamp.

Table 1. Values used for clamping pressure calculations.

Parameter	Value
d_o	54 *mm*
d_i	47.6 *mm*
d	50.8 *mm*
δ	0.1 *mm*
E_o	6 *MPa*
E_i	72,000 *MPa*
ν_o	0.35
ν_i	0.17
μ_s	0.64
A	8107.3 mm^2
I	0.001405 $kg\ m^2$
α	0.1742 rad/s^2

Using these basic values for calculation, the pressure applied by the clamp and the corresponding maximum transmittable torque are found using the following equations Budynas et al. (2011), Eq. 1 – Eq. 3, and results listed below in Table 2.

$$P = \frac{\delta}{\frac{d}{E_o}\left(\frac{d_0^2 + d^2}{d_0^2 - d^2} + \nu_0\right) + \frac{d}{E_i}\left(\frac{d^2 + d_i^2}{d^2 - d_i^2} + \nu_i\right)} \quad \text{Eq. 1}$$

$$F = \mu_s P A \quad \text{Eq. 2}$$

$$T = \frac{Fd}{2} \quad \text{Eq. 3}$$

Table 2. Results of clamping pressure analysis.

Parameter	Value
P_{clamp}	705.5 Pa
$T_{transmittable}$	0.093 Nm

A calculation was performed using dynamic principles to verify the transmittable torque would be enough to rotate the lamp. First the mass moment of inertia of the flashlamp was identified from the SolidWorks model, then a base rotational acceleration was used from the Thor Labs rotary table, both shown previously in Table 1. These values were multiplied together in accordance with Eq. 4, the result is displayed below in Table 3.

$$T_{Load} = I\alpha \quad \text{Eq. 4}$$

Table 3. Results of dynamic loading calculations.

Parameter	Value
T_{Load}	0.000115 *Nm*

It is observed that T_{Load} is less than $T_{transmittable}$ meaning the clamp will be able to transmit enough torque to rotate the flashlamp under the provided parameters. The calculated load is also less than the maximum continuous torque that the ThorLabs rotary table can provide. This analysis proves that the proposed clamp and rotary system will be able to hold and rotate the flashlamp without issue.

5.2 Cycle Time Analysis

To estimate the cycle time of the solution, several parameters are required. These parameters include the total length of the glass of flashlamp, l, diameter of the flashlamp, d, the horizontal/vertical field of views of the lens, $HFOV / VFOV$, the framerate of the camera, FPS, the maximum velocity for the ball screw, v_b, the maximum angular velocity of the rotary table, ω_r, allowable pixel blur, N_{pixel}, pixel size of camera, p, magnification of lens, M, and exposure time, t_{exp}. These parameters are given below in Table 4.

Table 4. Parameters used for cycle time calculations.

Parameter	Value
l	2000 *mm*
d	50.8 *mm*
$HFOV$	21 *mm*
$VFOV$	10.8 *mm*
FPS	35 *FPS*
v_b	150 mm/s
ω_r	25 deg/s
N_{pixel}	0.5 *pixels*

p	2.5 μm
M	0.5x
t_{exp}	20 ms

The cycle time is constrained by factors including the linear velocity of the rail, the angular velocity of the rotary table, the framerate of the camera, and motion blur. To image the entire flashlamp, the camera should process the length of the lamp, then the lamp will rotate to provide a new field of view. A vertical field of view calculation is performed to estimate rotations required to view the full surface area, a diagram is shown in Figure 56.

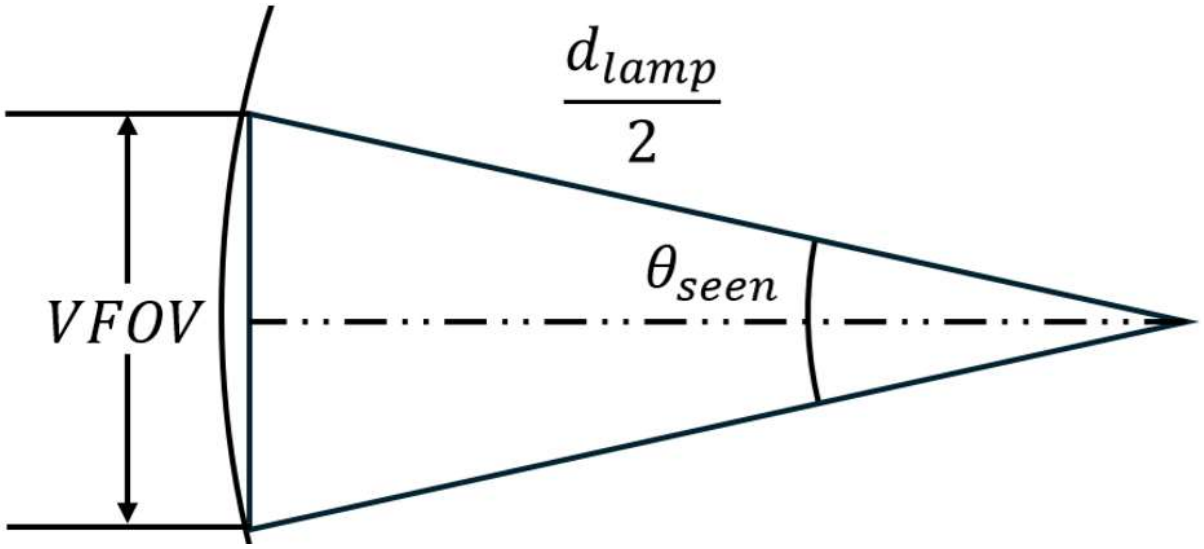


Figure 56. Angle of flashlamp seen by camera.

The number of times that the lamp has to rotate will be determined by the vertical field of view. By applying trigonometry, the angle of the flashlamp which the lens can see at a given time, θ_{seen}, can be found as shown in Eq 5.

$$\theta_{seen} = 2\sin^{-1}\left(\frac{VFOV}{d}\right) \quad \text{Eq. 5}$$

By dividing the circle by θ_{seen}, the total number of rotations required can be found, which is 15 rotations based on the lenses VFOV. θ_{seen} is around 24.5°, so the rotary table can be approximated to roll between rotations in about a second, based on ω_r. Thus, the estimated total time dedicated to rotation in a cycle to be 15 seconds.

The maximum possible velocities which the ball screw can translate are dependent on v_b, FPS and motion blurring. The manufacturer of the ball screw provided v_b, which has been tabulated previously as 150 $\frac{mm}{s}$. The maximum velocity that the camera can move while still imaging the entire lamp constrained only by FPS, v_{FPS}, can be found with Eq. 6.

$$v_{FPS} = (HFOV)(FPS) \quad \text{Eq. 6}$$

v_{FPS} is then found to be 735 $\frac{mm}{s}$, well above v_b. The final component affecting our ball screw velocity is motion blur. Eq. 7 gives the highest velocity, v_{blur}, that can be achieved for a desired amount of N_{pixel}.

$$v_{blur} = \frac{N_{pixel}p}{Mt_{exp}} \quad \text{Eq. 7}$$

N_{pixel} is chosen to be 0.5 as an unofficial standard for maximum allowable pixel blur for machine vision applications. t_{exp} is a parameter of the camera, which is freely programmable, so is chosen as 20 ms, as a lower t_{exp} would mean much heavier lighting requirements. Based on those parameters and the parameters of the camera and lens, v_{blur} is found to be 125 $\frac{mm}{s}$. Since it has the lowest maximum velocity, it will be the velocity which constrains the system. It is also desirable because it adds a factor of safety onto v_b, which improves the lifetime of the ball screw.

By approximating the velocity as being constantly v_{blur}, a simplification which gives a good ballpark estimate, the total cycle time, t_{cycle}, is estimated in Eq. 8. An additional t_{alg} is added onto this final time to account for any delay the software will take to process images, estimated to be 2 minutes. The result is provided in Table 5.

$$t_{cycle} = t_{alg} + t_{rot} + 15\left(\frac{l}{v_{blur}}\right) \quad \text{Eq. 8}$$

Table 5. Results of cycle time analysis.

Parameter	Value
t_{cycle}	6.25 *min*

t_{cycle} is observed to be lower than the would be nice time requirement of 15 minutes. This means that if any other constraints arise with cycle time, there is significantly more time to allocate to various aspects of a cycle.

5.3 Resolution Calculation

The most important parameter of the optical system is the number of pixels subtended by the characteristic length of a 0.3 *mm* defect (e.g. the diameter of a bubble), which will be referred to as n_{pixel}. n_{pixel} characterizes the resolution of the system; we can digitally zoom on our smallest defect and get a certain resolution. To calculate this value for any system, Eq. 9 is used.

$$\frac{n_{pixel}}{RES} = \frac{0.3\ mm}{FOV} \quad \text{Eq. 9}$$

RES and FOV can be either the vertical or horizontal number of pixels and field of views. By solving this equation for n_{pixel} using parameters of the chosen lens and camera, n_{pixel} is found to be ~70 pixels. Optics companies such as Keyence, Elementary, and UnitX cite 3 pixels as a

minimum for detection and 16 pixels as a minimum for categorization. So, the n_{pixel} of the chosen system is sufficient for detection and categorization.

6. Planning

A project schedule was developed at the start of the project using a critical path analysis method. This schedule was organized in a Gantt Chart format, allowing for workflow visualization, consistent progress monitoring, and working hour estimation. As the project timeline progressed, the schedule was updated to match real life actions as well as adjust future tasks based on new information. The scheduling was broken into semester one and semester two subsections, each with further subsections based on milestones. By factoring lead times into item procurement, required ordering dates can be assessed and planned for in advance to ensure minimal delays in assembly. Both semester Gantt Charts can be seen in Appendix Figure E1 and E2. Across the length of the project there is a planned total of 692 labor hours.

7. Budget

During the design phase, the project budget was initially set at $10,000. However, as the project scope became more clearly defined, additional purchases were required to meet the system requirements, increasing the total allocated budget to $16,061. A Chico State overhead cost of 15% is applied to the total budget which accounts for a $2409.15 cost. Following the build phase, the material cost was calculated from the bill of materials to be $12,926.17, which together with the overhead charge accounts for approximately 96% of the total budget. Figure 57 provides a cost breakdown for each project section. This left a remaining balance of $584 from the $16,061 allocated budget.

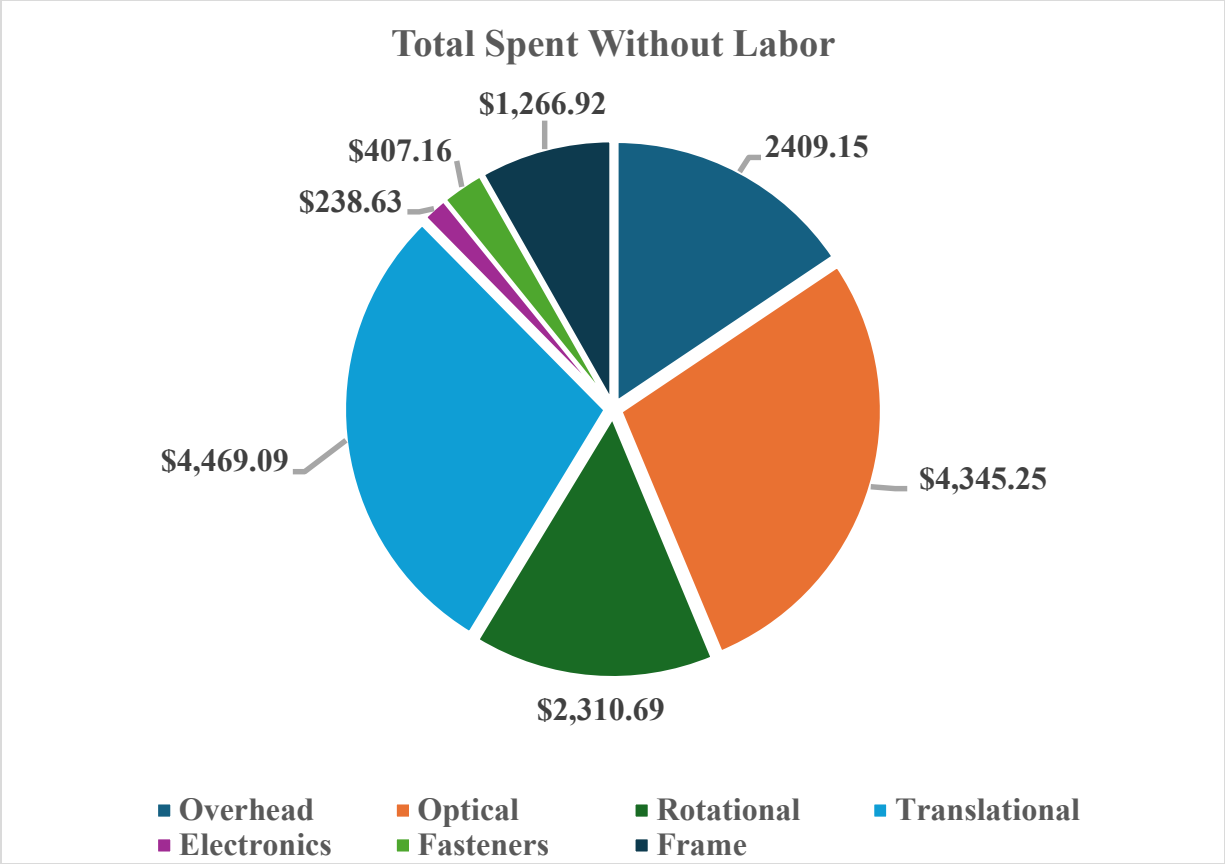


Figure 57. Total Cost without Labor.

In addition to material costs, the team completed a total of 692 labor hours, divided equally among all members. Using the United States median hourly rate for mechanical and mechatronic engineers, the estimated labor cost amounts to $68,119. The hourly rate for each person and the total labor cost calculation can be seen below in Table 6. The total project cost breakdown including labor can be seen in Figure 58.

Table 6. Estimated Labor Cost with Benefits and Overhead.

Engineer	Major	Hourly Rate	Time (hrs)	Estimate Labor Cost	Benefits and Overhead
Lucas McLaren	Mechanical	$52.92	173	$9,155.16	$16,204.63
Benjamin Nelson	Mechanical	$52.92	173	$9,155.16	$16,204.63
Ethan Waterman	Mechanical	$52.92	173	$9,155.16	$16,204.63
Samantha Paul	Mechatronics	$63.70	173	$11,020.10	$19,505.58
		Total Hours:	**692**	**Total Project Cost:**	**$68,119.48**

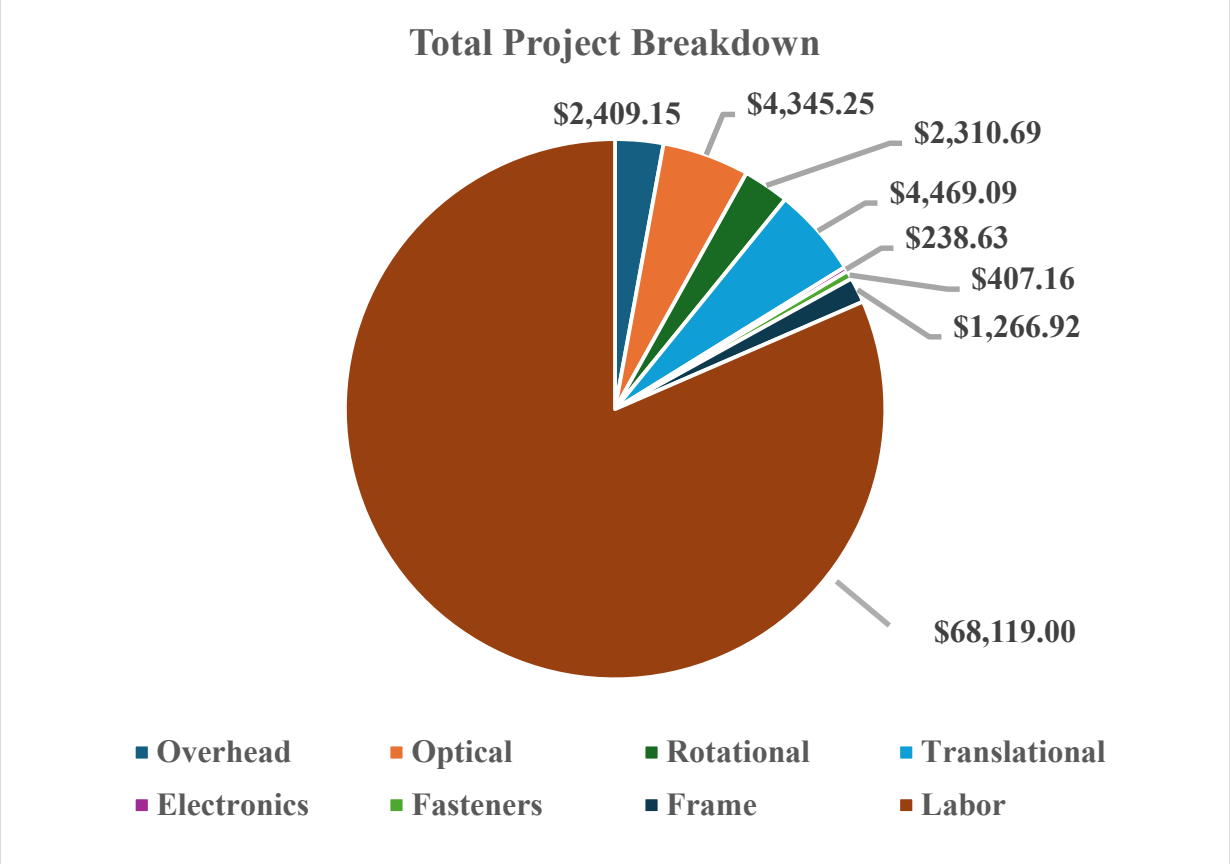


Figure 58. Total project cost breakdown.

The balance sheet for total cost broken up into overhead, purchased parts and labor can be seen below in Table 7. The total cost for the project including labor is $83,454.80.

Table 7. Balance Sheet of Total Cost.

Total Project Cost	
Parts:	$12,926.17
Labor:	$68,119.48
Overhead:	$2,409.15
Total:	**$83,454.80**

8. Justification of Solution

The success of the design is based on how many project requirements are completed by the end of the project term. This project consists of 11 must do, 5 should do, and 4 would be nice requirements. The requirements and proposed solutions are listed.

8.1 Must do requirements:

- Finding Defects that are 0.3 *mm* or larger and being able to identify 6 types:
    - The lensing system subtends 70 pixels for 0.3 *mm* ensuring detection and categorization.
- Ultraviolet, White light and eye safety:
    - This system has a main white light and a 395 *nm* UV light that is held above the lamp by adjustable pivot arms for the camera to better identify defects.
    - UV protective eyewear is provided alongside the project for eye safety.
- Live Feed, Manual Operation mode and LabVIEW Compatibility:
    - LabVIEW is the choice of software used for the project and within the program a manual operation mode is available.
    - The software and machine learning model we use is trained to identify at least 6 defect types with the ability to add more.
    - Live feed is shown during all system operations.
- The System must comply with LLNL Safety Standards.
    - All Parts designed and most purchased are within LLNL Safety Standards.
    - Additional inspection for select purchased parts will allow the system to be in full compliance with LLNL standards.

8.2 Should do and would be nice requirements:

- Metric Units and UL Listed Electronic Parts:
    - All parts designed and purchased are within Metric Unit standard.
    - Most electronic parts are UL Listed or capable of inspection.
- User Friendly Display and Aesthetically Pleasing:
    - The program has a user-friendly UI and is aesthetically pleasing.

9. Test Results and Interpretation

The preliminary test results shown in Appendix F show that all requirements tested were successfully achieved. Beginning with the cycle time testing, it was found that the cycle time falls within the requirement of sub 15 minutes, with an average cycle time of 9:31:37, standard deviation of 0.6 seconds, and an uncertainty of 0.2 seconds. This uncertainty is very small and can be accounted for in stopwatch measurement error.

The following test, rotational accuracy, also fell within the requirement range of 90% locational accuracy. The test proved that the rotation stage is over 99.9% accurate and will not be a source of locational uncertainty for the system.

The final test, translational accuracy, proved that the system accomplishes the locational accuracy requirement to the highest degree, with an average misalignment of only 0.1 *mm*, it provides an accuracy of over 99.9%. The mean misalignment can be attributed to measurement error as the measuring tape used had a measurement uncertainty of 0.5 *mm*.

After running these tests, the system has proven that it not only accomplishes the required task under the 15-minute requirement, but provides accurate locational data associated with found defects to over 99.8% accuracy.

10. Summary and Recommendations

The automated flashlamp inspection system provides a solution for achieving consistent, full-surface evaluation of flashlamps and tubes. The selected mechanical configuration integrates a rigid support frame, controlled rotary motion, and linear translation. It meets the functional requirements identified by the sponsor, with low mechanical complexity and is designed for a long lifespan. The electrical and control architecture supports synchronized motion and imaging, ensuring the system can ultimately support LLNL's inspection accuracy and repeatability objectives. The project's core engineering constraints are addressed, all must-do requirements are satisfied, and there is a clear path for future use at LLNL.

Following up on the project, the system is ready for use following installation and inspection at NIF. The system reduces technician manual labor in both time and physical intervention providing NIF superior inspection efficiency. There are several recommended design improvements to increase the quality of the system. Primarily, reprinting all 3D printed components with Polyether ether ketone (PEEK) for extended lifespan and quality. Implementing a cover on the front of the frame to improve image consistency and remove background color differences from defect images. A stronger light both white and UV may make defect images even clearer. Automating the movement of the secondary camera would decrease technician intervention further but may require an increased footprint. A more resolute primary and/or secondary camera would increase inspection capabilities but may require more data storage due to larger image sizes.

11. Impact and Contemporary Issues

The Automated Flashlamp Microscopic Flaw Inspection System provides a solution which streamlines a critical path in the research of fusion energy technologies. It will allow for testing and research to occur faster and decrease the timeline for fusion reactor power plants to become a reality. Fusion reactors are the one of the first true green energy solutions globally and will shape the energy sector across the world. Economically, the creation of consistent fusion reactors will lower energy prices for everyone and allow for further technological advances due to decreased energy investment costs. Environmentally, the development of fusion energy will allow for reduced fossil fuel requirements and assist in the decrease of greenhouse gas emissions. Socially, the reality of green energy will become a common-known fact and will shape the cultural context of fossil fuel dependency.

REFERENCES

Budynas, R. G., Nisbett, J. K., & Shigley, J. E. (2011). Shigley's mechanical engineering design. New York: McGraw-Hill.

APPENDIX

A – Project Requirements

Table A1. Must do requirements.

REQUIREMENT	SPECIFICATION	TESTING PROCEDURE
Find Defects	≥ 0.3 mm	Lamp Testing
Identify Critical defects	≥ 6 Types	Lamp Testing
Ultraviolet Light	100-400*nm*	Check Laser Specs
Broadband Light	400-700*nm*	Check Light Specs
Live Feed Display	---	Run Cycle, Check Feed
Manual Operation Mode	---	Run Cycle, Manually Operate
Accommodates Lamps/Tubes	---	Run Cycles with Tube/Lamp
Collect Data in LabVIEW	---	Program Inspection
Store Data in Exportable Format	---	File Inspection
120V AC Compatible	---	Plug into Outlet
Eye-safe	---	LLNL Safety Standards
Comply with LLNL Safety Standards	---	Consult Safety Standards
Run Time	< 30min	Run Full Cycle

Table A2. Should do requirements.

REQUIREMENT	SPECIFICATION	TESTING PROCEDURE
Identify Different Defects	≥10 Types	Lamp Testing
Metric Units	---	Inspection
UL Listed Electronics	---	Inspection
User Friendly Interface	---	Technician Inspection
Active Process Display	---	Inspection

Table A3. Would be nice requirements.

REQUIREMENT	SPECIFICATION	TESTING PROCEDURE
Identify All Defects	30 Types	Lamp Testing
Short cycle time	≤15 Minutes	Stopwatch
Automatic Microscope Pitching	---	Lamp Testing
Aesthetically Pleasing	---	Technician Inspection

B – Working Drawings

Following this page is a complete set of working drawings.

C – Pugh Matrices

Table C1. Rotation Pugh analysis.

Criteria	Weight	Hot Dog Roller		Lathe Chuck		Clamp	
		Base Score	Weighted Score	Base Score	Weighted Score	Base Score	Weighted Score
Cost Efficiency	3	4	12	1	3	3	9
Accuracy	2	1	2	2	4	5	10
Manufacturability	1	4	4	1	1	3	3
Ease of Use	4	4	16	1	4	4	16
Reliability	5	1	5	3	15	3	15
Total Weighted Scores:			**39**		**27**		**53**

Table C2. Translation Pugh analysis.

Criteria	Weight	Belt and Pully		Rack and Pinion		Ball Screw	
		Base Score	Weighted Score	Base Score	Weighted Score	Base Score	Weighted Score
Cost Efficiency	3	5	15	3	9	3	9
Accuracy	2	3	6	5	10	5	10
Manufacturability	1	5	5	4	4	5	5
Ease of Use	4	4	16	4	16	5	20
Reliability	5	1	5	5	25	5	25
Total Weighted Scores:			**47**		**64**		**69**

D – Electronics Schematics

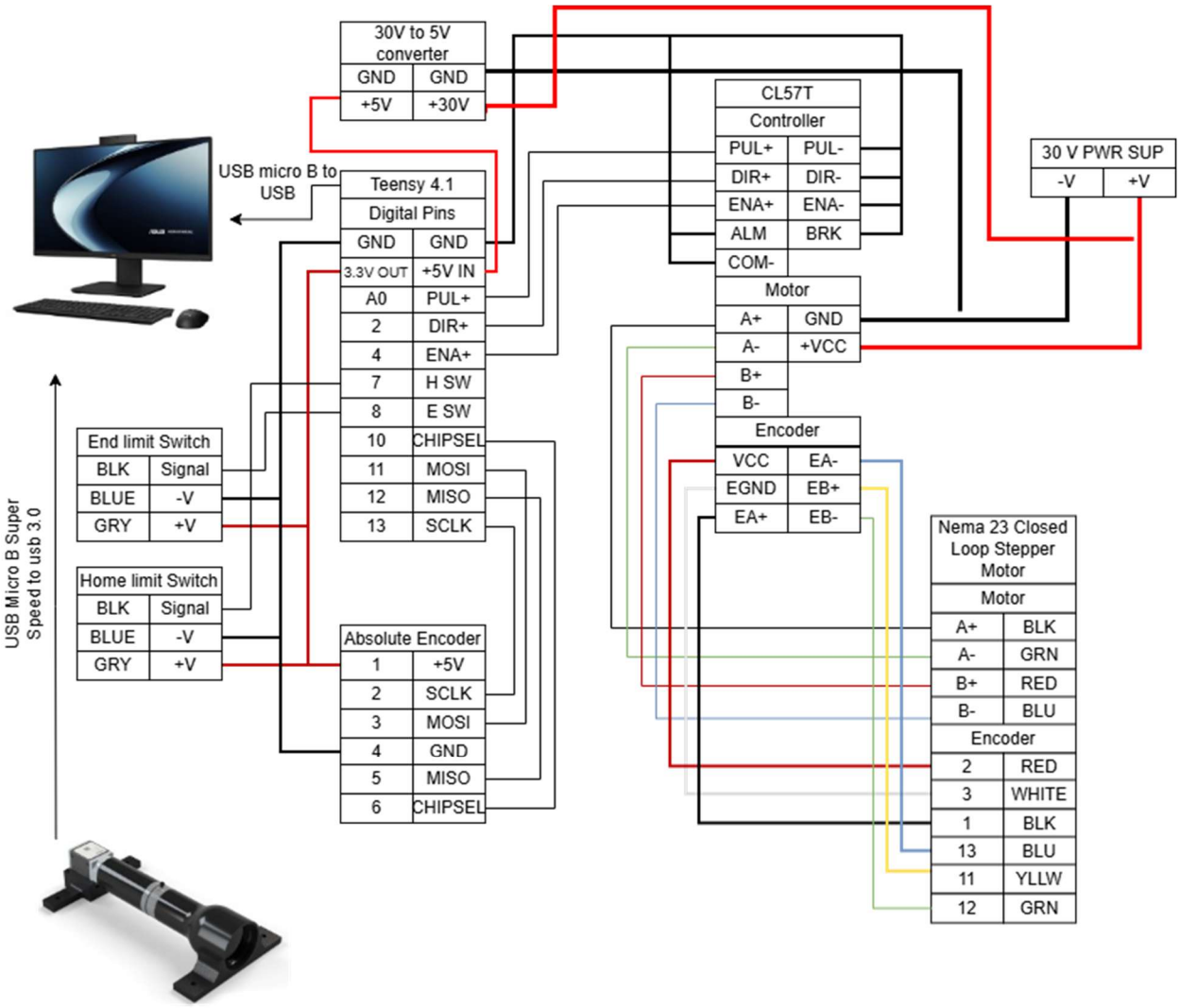


Figure D1. Wiring diagram for motion system.

E - Project Scheduling

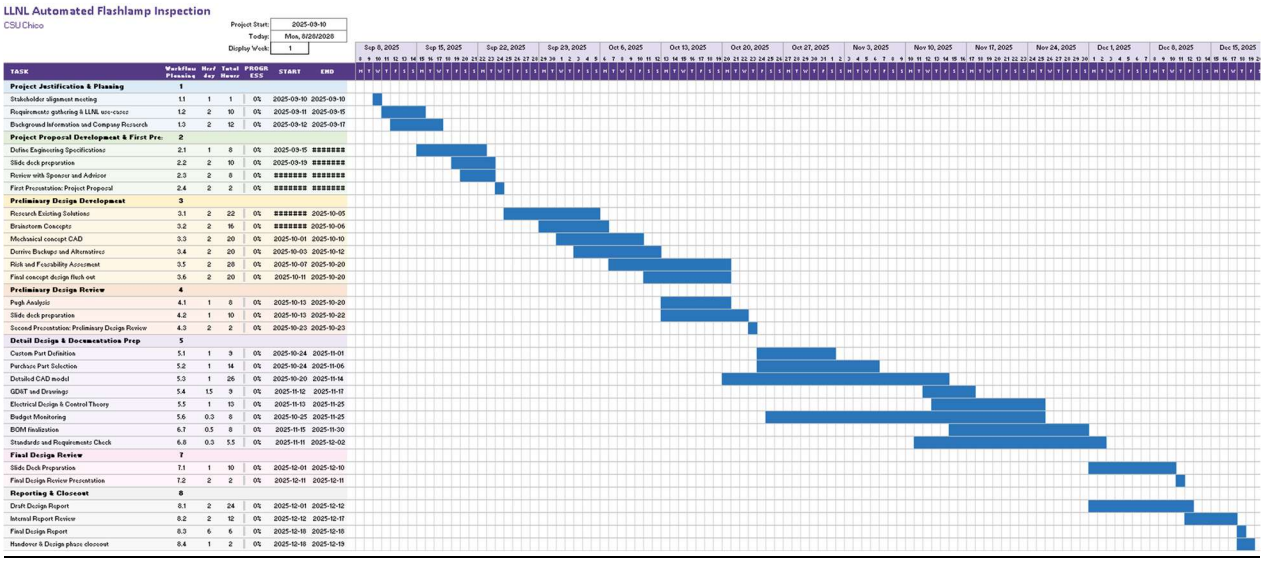


Figure E1. Semester 1 Gantt Chart

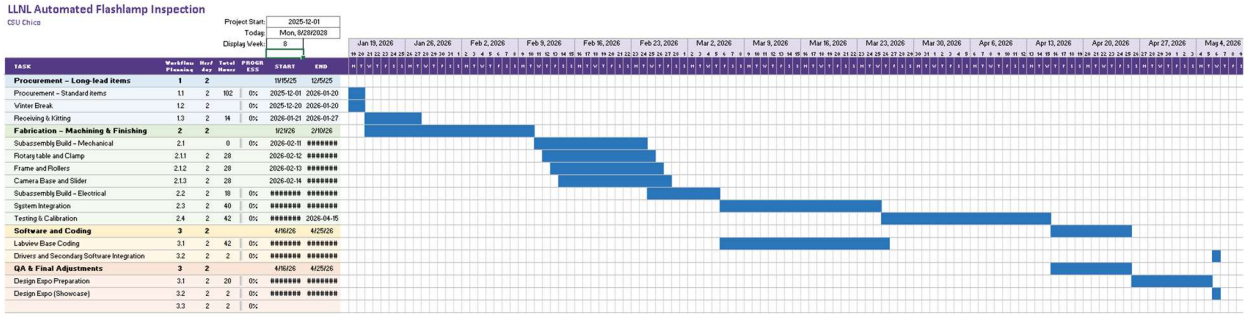


Figure E2. Semester 2 Gantt Chart

F – Test Procedures and Results

Following this page is a complete test procedure document and completed data sheets from preliminary testing

Test Plan - Cycle Time- Data Sheet

Lawrence Livermore National Laboratory

Data Collection

Trial	Stopwatch Time (mm:ss:ms)	Valid Cycle	Notes
1	9:31:12	yes	
2	9:31:86	yes	
3	9:32:02	yes	

Calculation

$$\bar{t} = \frac{1}{n}\sum_{i=1}^{n} t_i$$

Stopwatch Mean $\bar{t}$	9:31:37

$$S = \sqrt{\frac{\sum(t_i - \bar{t})^2}{2}}$$

Stopwatch S	0.60 ms

Additional Notes

Certification

Test Completed By	Date Completed	Time Completed	Location Completed
Lucas McLaren	5/11/2026	11:06 PM	Chico State LANG 122

Figure F-1. Cycle time preliminary testing data sheet.

Test Plan - Rotation Accuracy - Data Sheet

Lawrence Livermore National Laboratory

Data Collection

Control (Stage Alignment)	0

Input Angle	Stage Angle	Δθ (Input - Stage + Control)
30	30	0
60	60	0
90	90	0
120	120	0
150	150	0
180	180	0
210	210	0
240	240	0
270	270	0
300	300	0
330	330	0
360	360	0

Additional Notes
Kinematic motor controller read out accurate to 0.0005 degrees Physical rotation stage markings uncertainty 0.5 degrees

Calculation

$$\overline{\Delta\theta} = \sum \frac{\Delta\theta_i}{12}$$

Mean $\overline{\Delta\theta}$	0

$$S = \sqrt{\frac{\sum(\Delta\theta_i - \overline{\Delta\theta})^2}{12}}$$

Standard Deviation	0

Certification

Test Completed By	Date Completed	Time Completed	Location Completed
Lucas McLaren	5/11/2026	10:30:00 PM	Chico State LANG 122

Figure F-2. Rotation preliminary testing data sheet.

Test Plan - Translational Accuracy - Data Sheet

Lawrence Livermore National Laboratory

Datum	Known Distance (KD)	Measurement 1 (M_1)	Measurement 2 (M_2)	Measurement 3 (M_3)	Measurement 4 (M_4)	Measurement 5 (M_5)
A	500	500.5	500	500	500	500
B	1000	1000.5	1000	1000	1000	1000.5
C	1500	1500	1500	1500	1500	1500

Calculation

$$\Delta x_i = |KD - M_i| \qquad \overline{\Delta x} = \sum \frac{\Delta x_i}{15}$$

Mean Misalignment $\overline{\Delta x}$
0.1 mm

Additional Notes
Measuring tape uncertainty 0.5mm

Certification

Test Completed By	Date Completed	Time Completed	Location Completed
Lucas McLaren	5/11/2026	9:46 PM	Chico State LANG 122

Figure F-3. Translation preliminary testing data sheet